UNIVERSITY OF LJUBLJANA
BIOTECHNICAL FACULTY
ACADEMIC STUDY IN MICROBIOLOGY

Tajda BOGATAJ

USE OF THE BLACK SOLDIER FLY (*Hermetia illucens*) FOR PROTEIN PRODUCTION FROM ORGANIC WASTES AND ITS MICROBIOME

M.Sc. THESIS
Master Study Programmes: Field Microbiology

Ljubljana, 2018

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UPORABA ČRNE VOJAŠKE MUHE (*Hermetia illucens*) ZA PRIDOBIVANJE PROTEINOV IZ ORGANSKIH ODPADKOV IN ANALIZA NJENEGA MIKROBIOMA

MAGISTRSKO DELO
Magistrski študij – 2. stopnja Mikrobiologija

Ljubljana, 2018

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AB Larvae of black soldier fly are known biowaste reduction organisms. In this study, larvae were fed with three different substrates (chicken feed-control substrate, food waste and oil waste). Their growth, organic matter, development to prepupae stage, waste reduction and gut microbial community was observed. Oil waste was not suitable substrate for the larvae. On the other hand, food waste and chicken feed were appropriate substrates, which gave heavy and organic rich larvae/prepupae, stimulated larvae development to prepupae stage and reduced wastes. Gut microbiota of the larvae was not affected by substrate microbial community and by the time. Dominant phyla presented were *Proteobacteria*, *Firmicutes* and *Bacteroidetes*.

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AI Ličinke črne vojaške muhe (*Hermetia illucens;* BSF-black soldier fly) so znane razgrajevalke bioloških odpadkov. V tej študiji smo ličinke hranili s tremi različnimi substrati (kokošja krma-kontrolni substrat, ostanki hrane in ostanki olja). Opazovali smo njihovo rast, delež organskih snovi, njihov razvoj v stadij bube, sposobnost razgradnje odpadkov in njihovo mikrobno združbo prebavil. Rezultati so pokazali, da odpadki olja niso primeren substrat za gojenje ličink BSF. Ostala dva substrata (ostanki hrane in kokošja krma) sta bila primerna za gojenje *Hermetia illucens*. Ličinke so se razvile v težke in organsko bogate zrele ličinke sposobne razgradnje odpadkov. Ostanki hrane in kokošja krma sta spodbudili prehod ličink v naslednji stadij bube. Na mikrobioto prebavil ličinke mikrobiota substratov ni vplivala. Prav tako čas inkubacije ni vplival na mikrobno združbo v prebavilih. Prevladujoča debla so bila *Proteobacteria*, *Firmicutes* in *Bacteroidetes*.

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ABBREVIATIONS AND SYMBOLS

Ac	acetic acid / ethanoic acid
BCR	biomass conversion rate
BSF	black soldier fly (*Hermetia illucens*)
Bu	butyric acid / butanoic acid
COD	chemical oxygen demand
CORS	conversion of organic refuse by Saprophages
D	overall degradation
DM	dry matter
DNA	deoxyribonucleic acid
FAO	Food and Agriculture Organization of the United Nations
Fo	formic acid / methanoic acid
GIT	gastrointestinal track
H’	Shannon diversity index
HPLC	high-performance liquid chromatography
i-bu	isobutyric acid / 2-methylpropanoic acid
IPIFF	International Platform of Insects for Food and Feed
ITS	internal transcribed spacer
i-val	isovaleric acid / 3-methylbutanoic acid
Lac	lactic acid
LOI	loss of ignition
MR	material reduction
MSW	municipal solid waste
nMDS	non-metric multidimensional scaling
OM	organic matter
OTU	operational taxonomic unit
PAST	paleontological statistics
PCR	polymerase chain reaction
Pr	propionic acid / propanoic acid
RH	relative humidity
rRNA	ribosomal ribonucleic acid
SDS	sodium dodecyl sulphate
SOM	sample organic matter
UIBK	University of Innsbruck
Val	valeric acid / pentanoic acid
VFA	volatile fatty acid
VS	volatile solids
WC	water content
WRI	waste reduction index

1 INTRODUCTION

Modern society is facing many problems: increasing human population, expansion of large cities, higher demands for food and increasing amounts of wastes.

Municipal solid waste (MSW) is the waste generated from domestic, commercial and construction activities which are collected and treated by municipalities. Exponential population growth and urbanization, development of social economy, coupled with the improvement of the living standard, have resulted in an increase in the amount of MSW generation throughout the world (Karak et al., 2012). A predominant characteristic of MSW produced from households, small businesses and institutions, is the high fraction of organic, hence biodegradable matter (-biowaste) (Riuji Lohri et al., 2017).

Treatment of biowaste reduces public health hazards, and offers environmental and economic benefits by converting waste into a hygienic product, diverting it from disposal sites, and providing a source of income. There are four groups of biowaste treatment technologies: (i) direct use (direct land application, direct animal feed, direct combustion), (ii) biological treatment (composting, vermicomposting, black soldier fly treatment, anaerobic digestion, fermentation), (iii) physic-chemical treatment (transesterification, densification), and (iv) thermochemical treatment (pyrolysis, liquefaction, gasification) (Riuji Lohri et al., 2017).

Black soldier fly (BSF) treatment is an emerging technology in organic waste treatment. It involves the transformation of biowaste into insect protein and insect oil (Riuji Lohri et al., 2017). Larvae of black soldier fly, *Hermetia illucens* L. (Diptera: Stratomyidae) can consume a wide range of organic material. The prepupal stage of this insect can be harvest and use as a valuable nutritious feed for animal livestock (Nguyen et al., 2015). Grease extracted from larvae, prepupae and pupae present a source for biodiesel production and residual wastes can be used for sugar production (Li et al., 2011b).

1.1 AIM AND SCOPE

The aims of this thesis were (i) to test how BSF larvae grow on three different biowastes, (ii) whether biowaste type has any effect on the growth rate and organic matter content of larvae, and (iii) how the larvae bacterial gut microbiota is changing over the time and between different substrates.

We anticipate that the results will explain which substrate is best degraded by larvae and gives us the highest yield in larval biomass (more organic matter, bigger and heavier pupae). The microbial community data collected from larvae intestinal tract will

indicate, if the substrates/biowastes have an influence on the gut microbiota, how the core microbiome is shaped by the three different biowastes and if there are any specific species present just in the specific biowaste/environment.

2 LITERATURE REVIEW

2.1 BIOWASTE TREATMENT TECHNOLOGIES

Biowaste is comprised of food and kitchen waste, market waste, yard and park waste, residues from food and wood processing industries. Uncontrolled biowaste causes a considerable threat to public and environmental health. It induces odor, attracts insects, rodents and other disease vectors. Biowaste can generate leachate that may contaminate surface and groundwater supplies. Uncontrolled disposal of biowaste emits methane, a major greenhouse gas (Riuji Lohri et al., 2017).

With development on biowaste management, we are influencing on improving of municipal solid waste management. Besides reducing public health threats and environmental burdens, returning resource value of waste into the economy reflects the paradigm shift towards a circular economy focused on "closing loops" through recovery, while at the same time considering new business opportunities and economic growth (Riuji Lohri et al., 2017).

Biowaste treatment technologies are processes that convert discarded biowaste into new products with potentially generating some value. Treatment technologies for urban solid biowaste are grouped in four main group categories: (i) direct use, (ii) biological treatment, (iii) physic-chemical treatment and (iv) thermochemical treatment (Riuji Lohri et al., 2017).

For sustainable waste recycling, a supply of adequate waste materials as input and the market demand for the output products is required. Suitable markets may be animal husbandry (e.g. meat/fish production), agriculture (e.g. crops production) and bio-energy (e.g. transport fuel, cooking fuel/heat/electricity) (Riuji Lohri et al., 2017).

An ancient form of waste treatment is direct use of biowaste. It is associated with low costs and simplicity. Versions of direct use technology are (i) direct land application, (ii) waste fed directly to animals and (iii) direct open combustion. Crucial for the use of these techniques is the composition of the biowaste. Contamination can easily endanger human, animal and environmental health. Physic-chemical treatment summarizes conversion processes that are induced by chemical reactions (chemical process of transesterification for biodiesel production) or apply physical and mechanical force (physical densification process for the production of pellets and briquettes). Thermochemical conversion processes apply heat to induce chemical reactions as a way of extracting and creating energy carriers as products. Those processes are combustion, pyrolysis, liquefaction and gasification. Thermochemical conversion processes proceed faster than biochemical processes, but require significant energy input. Biological

treatment processes are controlled conversion of waste by living organisms. They always take place in a moist environment (all living organisms require water for survival). Biological treatments are divided into five groups: (i) composting, (ii) vermicomposting, (iii) black soldier fly (BSF) treatment, (iv) anaerobic digestion and (v) fermentation (Riuji Lohri et al., 2017).

2.1.1 BSF treatment

BSF treatment is an emerging technology in organic waste treatment. It involves the transformation of biowaste into insect protein and insect oil. Suitable waste sources for larvae composting are diverse and there is no general rule for the suitability of a waste source for BSF treatment. Black soldier fly larvae can be fed with food and market waste (Diener et al., 2011), animal manure (Li et al., 2011b), human excreta (Lalander et al., 2013; Banks et al., 2014) and fish waste (St-Hilaire et al., 2007). Larvae growth on highly cellulosic waste such as wood and dry leaves is slower. When using in larvae composting treatment it is best to add this substrate as a structure-forming agent. In an industrial BSF treatment facility, incoming waste has to be shredded to reduce particle size and the water content must be between 65 and 80% (Riuji Lohri et al., 2017).

The growth rate of BSF larvae, waste reduction and bioconversion rate depend on temperature (25-32 ℃) and moisture content of the feedstock. The BSF develops through six larval instars with the last larval stage (pre-pupae) crawling out of the moist feed source in search for a dry pupation site. Under controlled conditions (Gainesville house fly diet, 28℃, 75% RH) the total development from egg to adult lasts 20-35 days. The larvae can reduce the feedstock weight by 50-80% and convert up to 20% (on a total solid basis) into larval biomass within approximately 14 days (Diener et al., 2011). Space requirements depends on operational parameters such as density of larvae and feeding rate – (i) high waste reduction (high larval density and low feeding rate) or (ii) high biomass production (low larval density and high feeding rate) (Parra Paz et al., 2015). Feeding rates reported from the literature range from 1.9 kg/m^2 (Diener et al., 2011) to 9.8 kg/m^2 (Parra Paz et al., 2015).

Larvae and residuals are the main products of BSF technology. Defatted insects have similar protein content and amino acid profile as fishmeal and may thus replace fishmeal in animal feed. There were already some experiments, which showed good results when fed to fish, chicken or pigs. Other possible products are the production of biodiesel from larvae or the use of larval chitin and oil. The residue, on the other hand, still contains valuable nutrients and might be used as a soil amendment (Riuji Lohri et al., 2017).

International Platform of Insects for Food and Feed (IPIFF) is promoting insects as a source of animal proteins for human consumption and animal feed. The Food and Agriculture Organization of the United Nations (FAO) estimates that "global meat demand in 2030 will stand at 72% above the 2000 value" (Kanaly et al., 2010) and the aquaculture market is expected to grow 50% between 2010 and 2030 (Fish to 2030, 2013). The result is a big rise in demand for protein-rich feed. *Hermetia illucens* is one of the insects with big potential for use as a protein-rich feed. In areas without traditional histories of entomophagy and with food policies that prioritize risk avoidance, namely Europe, have more stringent rules about insects as a "novel food" that must be addressed before insects can be marketed as a human food (Wang and Shelomi, 2017). From 24 of May 2017, the regulation (Commission regulation (EU) 2017/893, 2017) identifies seven insect species "currently reared in the Union", including *H. illucens*, that fulfill the safety conditions for insect production for domestic and pet animal feed (Wang and Shelomi, 2017).

2.2 BLACK SOLDIER FLY

The black soldier fly, *H. illucens* (Linnaeus), is a common and widespread fly of the family Stratiomyidae, which comprises 2620 known and valid species (ITIS, 2017). Adult flies range in color from yellow, green, black or blue, with some having a metallic appearance. Many are mimics of other insects, like wasps and bees (Diclaro and Kaufman, 2009; Oliveira et al., 2015).

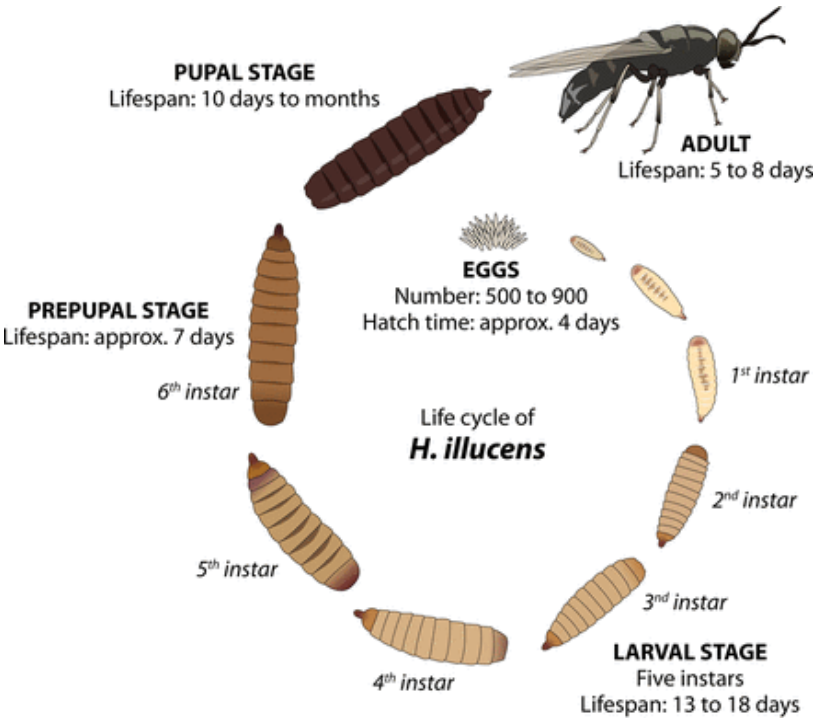

Figure 1: Life cycle of black soldier fly – *H. illucens* (Smet et al., 2018).

Taxonomic hierarchy of *H. illucens*:

- Kingdom: Animalia
- Phylum: Arthropoda
- Class: Insecta
- Order: Diptera
- Family: Stratomyidae
- Genus: *Hermetia*
- Species: *H. illucens* (ITIS, 2017)

BSF is native to the Neotropical region (Amazonia, Caribbean, Central America, Central Andes, eastern South America, Northern Andes, Orinoco and southern South America), but now it is spread across all the continents. Flies are present in Europe (in the Iberian Peninsula, in southern France, in Italy, Croatia, Malta, Canary Islands and Switzerland), in the Afrotropical ecozone, Australasian region, East Palearctic ecozone, Nearctic ecozone, North Africa and in the Oriental region (Diclaro and Kaufman, 2009; Oliveira et al., 2015; Fauna Europaea, 2016).

2.2.1 Description and life cycle

The black soldier fly, *H. illucens*, became one of the most important insect in the world of bioconversion. It is reared by multiple companies on an industrial scale of tones per week or month. Its advantage is short life cycle of 40-45 days (Smet et al., 2018) which contains four stages: (i) adult fly, (ii) eggs, (iii) larvae and (iv) pupae (Diclaro and Kaufman, 2009).

H. illucens adults have a wasp-like look and are black or blue in color. They do not need additional nutrition for survival and are not consider as a pest, because they do not enter houses and have a short life span (Nguyen et al., 2015). They have two translucent “windows” located in the first abdominal segment. Adults range from 13 to 20 mm in length, its antennae are elongate with three segments and legs have white coloration near the end of each leg. Two days after adults emerge from the pupal case, mating starts. A male black soldier fly utilizes lekking sites and intercepts a passing female in mid-air. They descend in copula (Diclaro and Kaufman, 2009; Nguyen et al., 2015; Diener et al., 2009).

The female deposits around 500-900 eggs in cracks and crevices near or in decaying matter (Smet et al., 2018). Eggs are oval shape, pale yellow or creamy white in color and 1 mm in length. After four days, the eggs hatch into larvae (Diclaro and Kaufman, 2009; Oliveira et al., 2015).

The larvae can reach 27 mm in length and 6 mm in width. They are whitish in color and have a small, projecting head with chewing mouthparts. In approximately 14 days, larvae pass through six instars, depending on the environmental conditions. During their development, they are insatiable feeders (Diclaro and Kaufman, 2009). They can reduce manure waste by 50%. BSF larvae are not vectors of disease and are capable of inactivating *Escherichia coli* and *Salmonella* (Nguyen et al., 2015; Erickson et al., 2004). They are known house fly (*Musca domestica*) breeding suppressor (Lohri et al., 2017). Larvae consist of approximately 44.0-45.2% protein and 31.4-33.0% fat (Nguyen et al., 2015; Diener et al., 2011). The last/sixth larval instar crawls from the feeding site to a dry area and initiates pupation. The exoskeleton (skin) becomes darker and the pupa develops within. The pupation process requires around two weeks (Diclaro and Kaufman, 2009).

2.2.2 Economic importance

Larvae are effective biowaste recyclers. Furthermore, they generate food source of sufficient quality for fish and other animals (Diclaro and Kaufman, 2009). Bacteria isolated from black soldier fly larvae can be used as a probiotic to enhance manure reduction and subsequently larval development (Yu et al., 2011).

Fat extracted from BSF larvae presents an alternative and non-feedstock source for bio-diesel production (instead of crop oil). Fuel properties of biodiesel produced from BSF larvae fats (density, viscosity, ester content, flash point and cetane number) are comparable to those of rapeseed-oil-based biodiesel (Li et al., 2011a).

The lauric acid (monolaurin) is one of the valuable product because of anti-microbial and anti-fungal properties. Sources of lauric acid are coconut milk, coconut oil, grated coconut and also larvae of black soldier fly (Leong et al., 2015).

Gut microbiota of BSF larvae is a potential source for new enzymes used in biotechnology, e.g. cellulose degradation gene (Lee et al., 2014). Recently, novel antimicrobial peptides from larvae of BSF were identified by screening, expression, purification and functional characterization of novel antimicrobial peptide genes (Park et al., 2014; Elhag et al., 2017). Their products exhibited diverse inhibitory activity on various pathogens, including Gram-positive bacterium *Staphylococcus aureus*, Gram-negative bacterium *Escherichia coli*, fungus *Rhizoctonia solani* Khün (rice)-10, and fungus *Sclerotinia sclerotiorum* (Lib.) de Bary-14 (Park et al., 2014; Elhag et al., 2017).

2.3 GASTROINTESTINAL TRACT OF BLACK SOLDIER FLY

For traditional farm animals is well establish that the microbiota in the gastrointestinal tract (GIT) significantly contributes to animal performance and health and that its composition is vary depending on the feed. This knowledge lacks for industrially reared insects (Smet et al., 2018).

The GIT of domestic animals harbors dense and complex microbial communities, composed of bacteria, protozoa, fungi, archaea and viruses. This microbiota is involved in (i) the digestion and fermentation of plant polymers, (ii) the synthesis of vitamins, (iii) the bioconversion of toxin components, (iv) the stimulation of the immune system, (v) the maintenance of gut peristalsis, (vi) keeping intestinal mucosal integrity and (vii) forming a barrier against colonization by pathogens. Gastrointestinal microbial community is affected by numerous factors like feeding practices, diet and farm management. Feeding efficiency, welfare and health of the animals can be affected by the alternations in the microbiota (Smet et al., 2018).

The GIT of insects also harbors high number of bacteria (ten-times more bacterial cells then insect cells). These bacteria serve several functions such as enabling digestion of specific compounds and distorting the sex ratio in some insects (Smet et al., 2018).

The GIT of *H. illucens* is divide in three parts, the foregut, midgut and hindgut, and all of them host different bacterial communities. The diversity of substrates used for growth of *H. illucens* larvae is remarkable and range from agricultural by-products such as potato steam peelings, bread remains or sugar beet pulp, seaweed, liver and cadavers to animal and human manure. The substrate impacts on the time needed for *H. illucens* to progress through its life-stages and on the nutritional composition of the larvae. It was shown, that larvae grown on a protein or a fiber-rich substrate develop much slower than larvae grown on a balanced diet of cereal processing leftovers. Reasons for substrate-dependent growth variations are (i) different composition of macronutrients between substrates, (ii) moisture level of the substrate and (iii) composition of substrate microbial communities (Smet et al., 2018).

Until now, few studies and observations about relationship between microbiota and growth of BSF were done. They illustrated that the microbiota does seem to play a crucial role during *H. illucens* growth. For example, larvae cannot grow on sterilized substrate. It needs to be inoculated with a small amount of untreated substrate from a previous rearing cycle in order to obtain proper zootechnical performance. Investigation of interactions between substrate-larvae community and its impact on larvae development require characterization of the microbiota in substrate and larvae during larvae rearing (Smet et al., 2018).

Jeon et al. (2011) published the first report of the bacterial community in *H. illucens* larvae. They characterized the gut microbial community during laboratory rearing on three different substrates (food waste, calf forage and cooked rice). Their goal was to investigate the role of the intestinal microorganisms in the remarkable catabolic abilities of the larvae (Jeon et al., 2011). The 16S rRNA gene sequences of the gut microbiome were analyzed using pyrosequencing. They found substrate-dependent shifts in the community structure of the GIT bacteria. The identified communities seemed unique compared to those of the intestinal microbiome from other insects. Overall diversity was linked to the nutritional complexity of the substrate. Food with more diverse nutrients gave higher microbial diversity (Smet et al., 2018). Bacterial community in the gut of larvae fed three different diets was mainly composed of four phyla with dissimilar proportions. Composition of larvae gut community is present on the Table 1. In the gut of larvae fed with food waste (from Korean restaurant), *Bacteriodetes* were the phylum with the highest content. Gut microbiota of larvae fed with cooked rice possessed mainly *Firmicutes* and *Proteobacteria*. Calf forage diet was a good substrate for *Firmicutes*, *Proteobacteria*, *Actinobacteria* and *Bacteriodetes*. *Proteobacteria* present in larvae fed with cooked rice and calf forage were mostly representatives of *Gammaproteobacteria*, while in larvae fed with food waste more than 65% of *Proteobacteria* belonged to the class of *Betaproteobacteria* (Jeon et al., 2011). In total, 36 bacterial species were identified in all three substrates. This point towards the existence of a unique core composition in the gut microbiota (Smet et al., 2018).

Table 1: Bacterial community of the BSF guts according to Jeon et al. (2011).

Substrate	Bacterial community
Food waste (from Korean restaurant)	*Bacteriodetes* (67.36%) *Proteobacteria* (18.85%) *Firmicutes* (9.4%) *Fusobacteria* (2.01%) *Actinobacteria* (1.92%)
Cooked rice	*Proteobacteria* (54.01%) *Firmicutes* (42.27%) Unclassified sequences (3.52%)
Calf forage diet (crude protein 15%, crude fat 15%, crude ash 1.5%, Ca 0.8%, P 0.8% and total digestible nutrients 67%)	*Firmicutes* (23.46%) *Proteobacteria* (31.08%) *Actinobacteria* (24.58%) *Bacteriodetes* (20.51%)

Two years after the first published research about *H. illucens* gut microbiota, Zheng et al. (2013) used 16S rRNA gene pyrosequencing to analyze the bacterial diversity during different life stages of BSF and to add weight to the existence of a core microbiota. His larvae were fed with Gainesville diet (20% corn meal, 30% alfalfa meal and 50% wheat bran, saturated with water) (Smet et al., 2018). In his study, bacterial DNA was isolated from the larvae, prepupae, pupae, adult and egg stage (they used the whole animal).

Bacteroidetes (42.0%) and *Proteobacteria* (33.4%) were the most dominant phyla associated with the black soldier fly across all life stages. In the prepupae (41.7%) and egg (54.0%) stages, *Proteobacteria* was the most dominant phylum. Dominant phylum in larvae (54.4%), pupa (46.3%) and adult (40.5%) stage was *Bacteroidetes*. Bacterial classes *Gammaproteobacteria*, *Sphingobacteria*, *Flavobacteria*, *Actinobacteria* and *Bacilli* comprised almost 78% of bacteria across all life stages. Bacterial orders *Sphingobacteriales*, *Flavobacteriales*, *Actinomycetales*, *Enterobacteriales*, *Burkholderiales* and *Bacteroidales* presented almost 78% of the bacteria across all life stages. Approximately 49% of all bacteria associated with the larval stage belong to *Bacteroidales* and *Clostridiales* whereas > 50% of all bacteria associated with pupal and adult stages belong to *Sphingobacteriales*, *Flavobacteriales* and *Actinomycetales* (Zheng et al., 2013). Results showed significant differences in bacterial community between life stages and significant clustering of bacterial communities associated with each of the life stages. Bacterial richness and diversity indices were calculated based on the Shannon diversity index (H'). At the species and genus level, H' was much higher in larvae, prepupae, pupae and adults than in eggs. At the phylum level, H' was similar in all life stages (Zheng et al., 2013).

Table 2: Number of the sequences, length from 200 to 541bp, presented in different stages of BSF, according to Zheng et al. (2013).

BSF life stage	Number of sequences
Larvae	476
Prepupae	1,080
Pupae	1,847
Adult fly	794
Eggs	124
SUM	4,321

Boccazzi et al. (2017) investigated the fungal community present in GIT of BSF. Mycobiota was investigated by culture-dependent approaches (microbiological analysis and molecular identification) and by culture-independent method (amplicons of the 5.8S-ITS rRNA region using 454 pyrosequencing). Larvae were reared in the laboratory on chicken feed solely for 17 or 21 days, or for 17 or 14 days followed by four or seven days respectively on vegetable waste. Results showed fungal community is impact by substrate alternations in a similar manner as the bacterial community. Fungal diversity in larvae GIT community increased when cultivated on substrate composed of various waste streams. Fungal communities tend to be dominate by a single genus depending on the feeding conditions than bacterial communities. In the experiment, *Pichia*, *Geotrichum* or *Trichosporon* dominated the community in three of the five feeding regimes tested. This can be because many yeasts are able to inhibit the growth of others using antagonistic mechanisms e.g. antimicrobial compounds (Boccazzi et al., 2017).

Extracts from the GIT of BSF showed high amylase, lipase and protease activities. Most of those activities are very high in the gut of the larvae. On the other hand, extracts from their salivary glands exhibited less than 10% of the total enzyme activity. GIT microbial community play an important role in aiding their host during the digestion of complex substrates by possessing metabolic properties that the insect lacks. Indeed, studies have shown that the metagenome of gut microbiota encoding a number of enzymes used in metabolic reactions. These microbial functions during digestion could be at the base of the observed substrate-dependent microbial community structure variations, as bacteria more adapted to digest a specific substrate will have a fitness advantage in the GIT over less adapted bacteria (Smet et al., 2018). On the other hand, insect by his self-produce antimicrobial peptides that can affect the microbial community structure (Elhag et al., 2017).

The metabolic capacities of *H. illucens* larvae and the observation of the high enzymatic activity in its GIT, present a bright source to find novel microbial enzymes such as proteases, cellulases, lipases, xylanases and pectinases that degrade organic compounds and might be of particular interest for industrial processes (Smet et al., 2018). With use of function-based metagenome screen technique for the larval gut microbiota, novel cellulose and an alkaline amylopullulanase was identified (Lee et al., 2014; Lee et al., 2016).

2.4 BIOCONVERSION OF DIFFERENT BIOWASTES BY BSF

Conversion of organic refuse by Saprophages (CORS) has been an issue over the last few decades. Vermicomposting is the most familiar example of CORS. It is conversion of organic waste by worms and microorganisms into black, earthly-smelling, nutrient-rich humus. Similarly, larvae of BSF was propagated as converter of organic waste, but also as a nutritious feed for chicken, pig breeding and aquaculture (Diener et al., 2009).

A CORS system combining waste treatment with concurrent protein production has to meet two objectives: (i) to achieve a high waste reduction efficiency and (ii) to yield a maximum biomass. To reach this aims the maximum amount of waste that can be process by larvae per day need to be determinate. Waste is consider fully processed when it has been subject to one larval digestion cycle. The resulting residues can be use in agriculture as nutrient-rich soil amendment. Digestion stabilizes the waste, reduces odor emissions as well as fungal and bacterial growth. For a CORS system to be economically viable, it must generate a suitable quality (protein and fat rich) and quantity of prepupae. Diener et al. (2009) wanted to show that there is a threshold in food supply, where larval dry weight does not increase further and larval development time is no longer reduce by an increase in food supply. They considered that a daily rate

of 100 mg chicken feed per larva met both the demand for a nutrient-rich prepupal output and high organic matter degradation within a short time (Diener et al., 2009).

Manure is the principal food for many insects in nature. Insect utilization contributes to natural recycling of nutrients and the produced insects serve as food for many larger animals. Insects convert residual manure proteins and other nutrients into their biomass, which is a high quality animal protein feedstuff. Larvae of BSF reducing pollution potential 50-60% or more with (i) nutrition incorporation and concentration from manure into valuable biomass (animal feedstuff) and (ii) larvae reduction of nutrient concentration and bulk of the manure residue. Because of its value ($0.50/kg or more), this feedstuff can be economically hauled significant distances (unlike manure) to relieve local nutrient overloads. Also, while larvae living in the manure they aerate and dry it, reducing odors. Maggots modify the microbial communities of manure, potentially reducing harmful bacteria (Erickson et al., 2004). The high-value insect feedstuff, reduction of the manure mass, moisture content, offensive odor and pollution potential are the returns for good management of such a system (Newton et al., 2005).

Newton et al. (2005) tested the suitability of swine manure as a substrate for BSF for successful manure management. In his small-scale system for digestion of swine manure, manure mass was reduced by 56%, while the concentration of most elements and nutrients were reduced for 40 to 55%. Nutrient analyses and feeding studies indicated that dried BSF prepupae grown on swine manure solids have value as a feedstuff, particularly for aquaculture. In a pot study, plant growth increased when the digested manure residue was add to either a clay soil or clean sand (Newton et al., 2005).

Li et al. (2011) investigated the hydrolysis of dairy manure after treatment with *H. illucens*. BSF larvae were expose to dairy manure for 21 days. The grease was extract from dried BSF and used as the material for biodiesel and the digested dairy manure hydrolyzed into sugars, which can be further use. In his study, approximately 1248.6 g fresh dairy manure were converted into 273.4 g dry residue by 1200 BSF larvae in 21 days. Approximately, 15.8 g of biodiesel were gained from 70.8 g dry BSF and 96.2 g sugar were obtained from the digested dairy manure (Li et al., 2011b).

Diener et al. (2011) further explored waste reduction capabilities of BSF larvae. They evaluated the feasibility of BSF larvae to digest and degrade mixed municipal organic waste in a medium-scale field experiment in Costa Rica and explored the benefits and limitations of this technology. Larvae were bred in “larveros” (80 cm x 200 cm x 30 cm), built with zinc-coated sheets. Organic waste was generate by the residents of the EARTH university campus, and mixed thoroughly to achieve a certain level of homogenization. Under favorable conditions, prepupae production of 252 g/m^2 day was

achieve. Waste reduction ranged from 65.5 to 78.9% depending on the daily amount of waste added to the experimental units and presence/absence of a drainage system. Three factors were identified to strongly influence larval yield and waste reduction capacity, (i) high larval mortality due to elevated zinc concentrations in the waste material and anaerobic conditions in the experiment trays; (ii) lack of fertile eggs due to zinc poisoning and (iii) limited access to food from stagnating liquid in the experimental trays (Diener et al., 2011).

Nguyen et al. (2015) cultivated larvae of *H. illucens* on five different substrates/biowastes ((i) control poultry feed, (ii) pig liver, (iii) pig manure, (iv) kitchen waste, (v) fruits and vegetables and (vi) rendered fish) and evaluated the uptake and reduction ability of black soldier fly larvae. Kitchen waste was the substrate with the greatest fat and calorie content. Comparing with the other used substrates (excluding fish), kitchen waste had the greatest mean rate of reduction per day and produced the longest and heaviest flies (prepupae). Fish had similar fat and calorie content to kitchen waste, but BSF larvae consumed fish at slower rate than kitchen waste. Nguyen et al. (2015) suggested, fish rendering might have contained an accumulation of heavy metals. It is known that *Drosophila melanogaster* larvae avoid foods with heavy metal contaminations when given a choice, as it interferes with larvae development and adult survival. When forced to consume a resource with heavy metal contamination, *D. melanogaster* larvae consume less as compared with diets with no heavy metals. That may be the same case with *H. illucens*. Fruit and vegetables was the substrate which was best consummated, but the rate of waste consumption was lower compared to standard poultry feed. This is because larvae fed fruit and vegetables took the longest to develop to the wandering stage. Manure had the lowest rate of waste reduction, possibly because it contained the lowest amount of energy and fat. The total amount of manure waste reduction was for 44% greater than that determined for the standard poultry feed. In comparison, the amount of waste reduced in kitchen waste (67.9%), fish rendering (74.2%) and fruit and vegetables (98.9%) was greater than standard poultry feed. With those results, Nguyen et al. (2015) showed that there is great promise for using BSF as a potential agent for waste management.

The exploitation of fast organic solid waste degradation by BSF larvae encouraged Parra Paz and her group (2015) to investigate the effects of larval density and feeding rate on the bioconversion of organic solid waste. There results showed, both variables have a significant influence on the bioconversion process, with larval density as the most influential element. Ideal conditions were determined, within the experiment's range, to be a larval density of 1.2 larvae/cm^2 and a feeding rate of 163 mg/larva day (dry base) which produces up to 1.1 kg/m^2 day of larval compost and 59 g/m^2 day of larval biomass, on dry base. In order to generate the highest amount of biomass, the process tolerates larval density values up to 5 larvae/cm^2 without significant influence

on the process as long as it is provided with a feeding rate no higher than 95 mg/larva day (dry base) (Parra Paz et al., 2015).

BSF larvae fed on organic waste have high protein and fat content. Because organic waste often contains persistent pollutants, such as heavy metals, that may accumulate in the larvae and prepupae of BSF and consequently in the food chain, its use as feed is questionable. Diener et al. (2015) fed BSF larvae with chicken fed spiked with heavy metals (cadmium, lead and zinc at three concentrations each) to examine the extent of metal accumulation in the different life stages and the effect of heavy metal concentration in the feed to the life cycle determinations of the flies. The study showed that concentrations of lead and zinc in larvae or prepupae remain below the initial amounts in the food. The accumulation factor of zinc in prepupae decreased with increasing zinc concentration in the feed. Furthermore, the three heavy metal elements examined had only minor effects on the development of the black soldier fly even at very high concentrations. Since cadmium accumulated in the prepupae, it could potentially limit the use of prepupae in the production of animal feed. In the case of lead and zinc, concerns about the use of prepupae in animal feed are less critical (Diener et al., 2015).

Lalander et al. (2016) investigated the accumulation of pharmaceuticals and pesticides in BSF treatment technologies. The study evaluated the fate of three pharmaceuticals (carbamazepine, roxithromycin, trimethoprim) and two pesticides (azoxystrobin, propiconazole) in a fly-larvae composting system and in the control treatment with no larvae. Dog food was use as the food substrate, as it is comparable to food waste in terms of nutrient composition and fat/energy content. The substrates were inoculate with the selected pharmaceuticals and pesticides. Results showed half-life of all five substances was shorter in the fly-larvae compost (<10% of control) and no accumulation was detected in the larvae. They believe that fly-larvae composting could thus impede the spread of pharmaceuticals and pesticides into the environment (Lalander et al., 2016).

3 MATERIAL AND METHODS

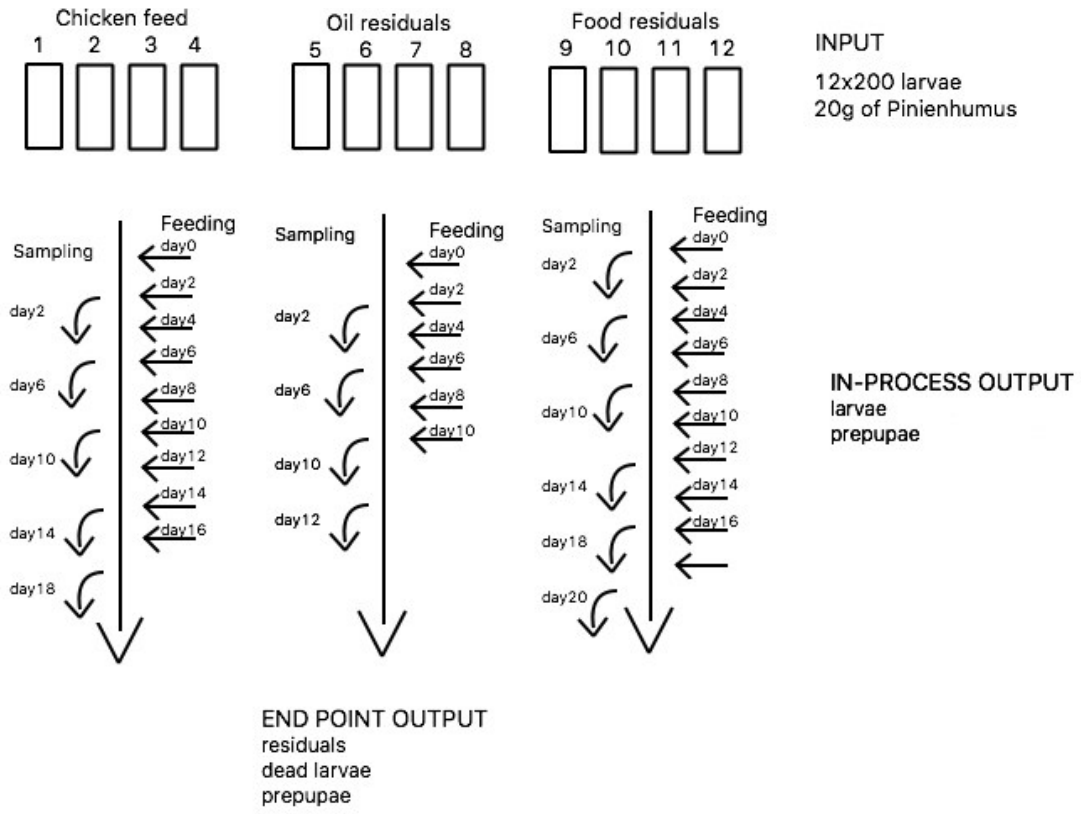

Figure 2: A schematic presentation of the feeding experiment design conducted in this study (Pinienhumus – *Pinus pinea*).

Feeding experiment, performed in this thesis, contained three different feeding regimes for the larvae of *H. illucens*. Each feeding conditions contained specific substrate (chicken feed, oil waste, food waste) and was conduct in four replicates. In the Figure 2, schematic representation of the experiment is present. In each of the twelve boxes, 200 larvae hatched within two days were put. Larvae in the boxes 1, 2, 3 and 4 were fed with chicken feed (Chicken feed pellets (Grünes LegeKorn Premium, Landwirtschaftliche Genossenschaft Klagenfurt, Austria)) every second day (feeding day 0, day 2, day 4, day 6, day 8, day 10, day 12, day 14, day 16). In the time of incubation, sampling of the larvae and prepupae were done on day 2, day 6, day 10, day 14 and day 18. At the end of incubation (when all larvae became prepupae/died), residues and the rest of the larvae and prepupae were collected. BSF in the boxes 5, 6, 7 and 8 were fed with oil waste every second day (day 0, day 2, day 4, day 6, day 8, day 10). In the time of incubation larvae were sampled (day 2, day 6, day 10, day 12) and dead larvae were removed. At the end of day 12 (all the larvae were dead), residuals were collected. *H. illucens* in the boxes 9, 10, 11 and 12 were fed with food waste every second day (day 0, day 2, day 4, day 6, day 8, day 10, day 12, day 14, day 16, day 18).

Larvae were sampled and prepupae were removed at day 2, day 6, day 10, day 14, day 18 and day 20. At the end of the feeding experiment, (all larvae developed in prepupae), residuals were collected. In-process and end-point output material were further analyzed.

3.1 MATERIAL

Table 3: Working organisms and substrates used in the experiment.

Organism/substrate	Description
Hermetia illucens	Larvae of black soldier fly obtained from Hermetia Baruth GmbH (Baruth Germany)
Gut microbiota	DNA extracted from BSF larvae gastrointestinal tract
Chicken feed	Chicken feed pellets (Grünes, LegeKor Premium, Landwirtschaftliche Genossenschaft Klagenfurt, Austria)
Food residuals	Food waste collected from biological waste bin (Mensa Technik Innsbruck, Technikerstrasse 13, Innsbruck) on 15.2.2017
Oil residuals	Waste oil, water, cooking, cleaning and grilling residuals were collected in the UIBK foodwaste collection tank (Mensa Technik Innsbruck, Technikerstrasse 13, Innsbruck); after layer making, upper two layers were collected (9.2.2017)
Pinus pinea	Pine humus (Dehner Terra, Rain, Germany)

Table 4: Cultivation, sampling, gut extraction and other laboratory equipment.

Cultivation	- climate incubator (25°C, humidity level 60%, 12 hour day/night cycle) (Fitotron® SGC 120 incubator (Weiss Technik UK, Leicestershire, United Kingdom)) - black plastic boxes (180 × 120 × 80 mm) - white plastic cups (50 mL) - wood shavings (Dehner Terra, Rain, Germany) - net (fiberglass; 150 × 150 mm, mesh size 2 × 2 mm) - rubber band transparent polypropylene cages (390 × 280 × 280 mm) with a net (fiberglass; 200 × 300 mm, mesh size 2 × 2 mm) integrated into the lid of each cage - wet sponges - 12 white incubation boxes (200 × 90 × 80 mm) - toothpick - spoon - balance - measuring cylinder - blender - electric grinder
Sampling	- petri plate - crucible - plastic bags - freezer - tweezers
Gut extraction	- micro centrifuge tubes - laminar flow cabinet - electrical burner - scalpel

continued

continuation of Table 4.

Laboratory equipment	- centrifuge 5804 R (Eppendorf) - vortex genie 2 (Scienctific industries) - scales (Mettler AE 166; DeltaRange®) - muffle furnace (Carbolite, CWF 1000) - desiccator - filter papers MN 615 ¼, 2r= 150 mm (Macherey-Nagel) - NanoDrop 2000c Spectrophoyometer (Thermo Scientific) - HPLC (Fast Fruit Column (Phenomenex, Germany) at 25°C; time program: (i) 0.25 mL/min 20 min, (ii) 1.0 mL/min 10 min, (iii) 1.0 mL/min 15 min; quantification and qualitative calibration: Volatile Acid Standard Mix (Sigm, Germany; 46975-U, Lot: LB 98693); internal standard: phenoxyacetic acid (Sigma, 158518)) - Leco TruSpec CHN Elemental Determinator (Leco, St Joseph, MI, USA) - pH meter (Metrohm) - electrophoresis (1% agarose gel, voltage of 100 V, 12 min) - spectrophotometer U-2001 (Hitachi) and NEC Pinwriter P2X (Printer)

Table 5: Chemicals, buffers, reagents and kits.

Chemicals	- $C_6H_5Na_3O_7$ * $2H_2O$ (Merck) - sodium carbonate (Roth) - SDS (Roth) - sodium hydroxide (Merck) - copper(II) sulphate pentahydrate (Roth) - Folin-Ciocalteus Phenolreagent (Merck) - D(+)-Glucose monohydrate (Merck) - 4-Hydroxybenzhydrazide (Sigma) - sodium sulphide (Sigma) - tri-sodium citrate dihydrat (Merck) - calciumchlorid-Dihydrat (Merck) - agarose (Sigma)
Buffers and reagents	- 0.5 x TAE buffer - 1M Na_2SO_3 - 0.5M Sodium citrate - 5M NaOH - 0.2M $CaCl_2$ - 1M KCl - Folin and Ciocaltes' reagent - copper sulphate reagent - alkaline copper reagent - Gene Ruler Express DNA Ladder (0.5 μg/μL) - GelGreen Nucleic Acid Strain, 10000x DMSO
Kits	- NANOCOLOR® CSB 1500 /COD/DCO/DQO (Macherey-Nagel, 2017b) - NANOCOLOR® Ammonium 50 (Macherey-Nagel, 2017a) - NucleoSpin® Soil (Macherey-Nagel, 2016)

3.2 METHODS

3.2.1 *H. illucens* cultivation

The cultivation of black soldier flies took place in a climate incubator (25°C, humidity level 60%, 12 hour day/night cycle) at the Department of Microbiology, University of Innsbruck. Adult flies were kept in rubber band transparent polypropylene cages with a net integrated into the lid of each cage. In the box, a test tube filled with water and sealed with cotton wool served as water source and small wet sponges were available for egg deposition. After oviposition, eggs were collected, using toothpicks and transferred into smaller black boxes covered with lids in which nets for aeration were incorporated. In those boxes, autoclaved pine humus was used as litter to cover up the sub-surface. Freshly hatched larvae were fed with chicken food (food:water=40:60 fresh weight) and the boxes were moisturized by adding water. When larvae transitioned to prepupal stage, prepupae were transferred into white plastic cups (2cm of pupae/cup), covered with wood shavings and a net. The flies emerging from the pupae were collected and transferred into rubber band transparent polypropylene cages with a net integrated into the lid of each cage where they continued the egg-laying process (Heussler et al., 2018).

In the feeding experiment, 2400 (12x200) individual larvae of *H. illucens* were used. Eggs were collected on 7.3.2017. They hatched between 10.3. and 11.3.2017 and were further fed with chicken food. On 17.3.2017, the feeding experiment started and larvae were exposed to three different substrates.

3.2.2 Characterization of substrates and residuals

Sample preparation for COD, ammonium content, protein and reducing sugar content were conducted as following. The 10 g aliquots of samples of either substrate or residual after incubation were added into 25 mL of deionized water. The mixture was mixed using vortex for few seconds, incubated for 30 min at room temperature and centrifuged at 25℃ and 12000×g for 30 min. After centrifugation, samples were filtered using filter paper (Macherey-Nagel, 2r= 150 mm).

Samples for pH, dry matter, volatile solid, HPLC, total C and N and fat content were conducted differently, as describe below.

3.2.2.1 Active and passive pH

The 10 g of substrate (chicken food, food residuals, oil residuals, autoclaved pinienhumus, residuals) were added to 25 mL of deionized water and mixed using

vortex. After 1 hour, the active pH in the mixture was measured with a pH meter (Metrohm).

For passive pH, 1 M KCl was prepared and its initial pH was determined. 10 g of substrate were added to 25 mL of 1 M KCl and mixed using vortex. The active pH was measured after 1 hour (pH meter, Metrohm) (Kassem and Nannipieri, 1995).

3.2.2.2 Water content and organic matter

Samples (larvae, pupae, chicken food, food residuals, oil residuals, pine humus, residuals) were weighted in a Petri dish and oven-dried (105°C) for at least 24 h. After this time, dried samples were re-weighted and total solid/water content was determined (1).

$$\text{WC}\ [\frac{\text{g H2O}}{\text{g dry sample}}] = \frac{(\text{sample fresh weight} - \text{sample dry weight})}{\text{sample dry weight}} \qquad \ldots(1)$$

Dried samples were homogenized and weighted into porcelain crucible for subsequent ignition in a muffle furnace (Carbolite, CWF 1000) at 550°C for 5h. After cooling the samples to room temperature in a desiccator, the porcelain crucibles were re-weighted and volatile solids were determined (2).

$$\text{SOM-LOI}\ [\frac{g\ OM}{g\ dry\ sample}] = \frac{(sample\ weight - ash)}{sample\ dry\ weight} \qquad \ldots(2)$$

3.2.2.3 COD (chemical oxygen demand)

Supernatant after filtration was diluted in dH_2O and COD was measured using a Nanocolor COD 1500 kit (MACHERNEY-NAGEL GmbH & Co. KG.) according to manufacturer's instructions.

3.2.2.4 Ammonium content

Preliminary prepared samples were diluted and ammonium content was measured using a Nanocolor Ammonium 50 kit (MACHERNEY-NAGEL GmbH & Co. KG.) according to manufacturer's instructions.

3.2.2.5 Total C and N

Samples (substrates, residuals) were oven-dried at 105°C for 24 h. After drying, dry samples were pulverized with a mortar and measured with Leco TruSpec CHN Elemental Determinator analyzer (Leco, St Joseph, MI, USA) according to manufacturer's protocol.

3.2.2.6 Protein content – Lowry

Preliminary prepared samples were diluted and measured using Lowry (Alkaline Copper Reduction Assay) assay according to Noble and Bailey (2009). Standard curve with equation A750 = 0.0019*protein conc.; $R^2 = 0.979$ is presented in Annex A.

3.2.2.7 Reducing sugar content

Preliminary prepared samples were diluted and measured using PAH BAH (p-hydroxybenzoic acid hydrazide) assay according to Lever (1977). Standard curve was prepared within the experiments conducted in this study and the equation $A_{420} = 0.0336$*sugar conc.+1.3084; $R^2 = 0.96005$ is presented in Annex B.

3.2.2.8 Fat content (unbound fat)

Samples (substrate, residual) were weighted into 15 mL plastic tubes and 5 mL of water were added. The tubes were centrifuged at 11000 × g (max speed) for 20 min or until phase separation was visible. The liquid phase was transferred in a petri dish and oven-dried at 105°C for 24 h. After drying, the petri dish was re-weighted and the fat content was calculated (3).

$$\text{Fat (\%)} = \frac{(tara + liquid\ phase) - (tara + after\ drying)}{mase\ of\ substrate\ befor\ adding\ water} * 100\% \qquad \ldots(3)$$

3.2.2.9 HPLC

Residuals and substrates were tested for the concentration of volatile fatty acids. The 1 g aliquots of sample were weighted in sterile 2 mL tubes. 1 mL of autoclaved water was added and tubes were centrifuged at 15000g for 15min. The liquid phase was transferred into a HPLC-vial and fatty acids were measured as described before (Wagner et al., 2017).

3.2.3 Calculation of mass balance

The percentage of biomass conversion rate (BCR) on a total solids basis was calculated (4).

$$\text{BCR}(\%)=\left(\frac{Mpp-Ml}{Fin}\right) * 100 \qquad \ldots(4)$$

M_l is the dry weight of the larvae inflow and M_{pp} the dry weight of the prepupae outflow. This is divided by F_{in}, the dry weight of the inflow material (mg) (Dortmans, 2015).

The percentage of material reduction (MR) on a total solid basis was calculated as (5),

$$\text{MR}(\%)=\left(1-\left(\frac{Mout}{Min}\right)\right) * 100 \qquad \ldots(5)$$

where M_{out} is the total solids of the material outflow and M_{in} the total solids of the material inflow. The outflow material was made up of the residue, not taking into account the residue that is recycled in the system and the dry weight of the sample. The material reduction is the amount of organic that either has been transformed into volatile compounds, such as CO_2 or NH_3, by biological processes, or is stored as biomass in the adult larvae that are removed from the system as prepupae (Dortmans, 2015).

For comparison, another two equations were used: (6) waste reduction index (WRI) and (7) overall degradation (D) (Diener et al., 2009). WRI was calculated using the overall degradation (D) divided by the number of days the larvae fed on the material.

$$WRI\left(\frac{g}{day}\right) = \frac{D}{t} \times 100 \qquad \ldots(6)$$

$$D = \frac{(W-R)}{W} \qquad \ldots(7)$$

For D calculation, data of the total amount of the organic material applied during the time t (W) and the residue after time t (R) were needed (Diener et al., 2009).

3.2.4 Feeding and sampling procedure

Twelve white boxes with closely sealable lids with incorporated aeration nets were prepared and weighted. In each box, 20 g of pine humus were added and the box was weighted again.

All substrates were mixed/homogenized, separated in bags and freeze until further use.

At the beginning of the experiment (day 0), 200 larvae were put in each box (box was weighted, to obtain larval weight). Boxes 1-4 were fed with chicken feed, boxes 5-8 with oil residuals and boxes 9-12 with food residuals. The amount of added feed and water was calculated according to the organic content of chicken feed (100 mg chicken feed/larvae day). Quantities used are present in the Table 6. Every second day (day 2, day 4, day 6, day 8, day 10, day 12, day 14, day 16, day 18) boxes were weighted, ten larvae were random removed from the box, weighted and put back into the box together with the respective amount of feed. If prepupae were present in the box, they were removed as well, weighted and DM and VS were determined. At days 2, 6, 10, 12 (boxes 5-8), 14, 18 and 20 (boxes 9-12), five larvae were used to determine DM and VS and another five larvae were put in a bag and freeze for subsequent gut extraction. In the boxes 5-8, dead larvae were detected and removed from the boxes.

At the end of the experiment, residuals were mixed, packed in bags and freeze for residuals analysis.

Table 6: Amount of substrate and water for larvae per day used in the feeding experiment.

Substrate	Mass of substrate/larvae per day (g)	Volume of water/larvae per day (mL)
Chicken feed	0.10	0.07
Oil waste	0.07	0.11
Food waste	0.17	0.00

3.2.5 Gut extraction

An electrical burner, three glass petri plates, ethanol, two tweezers, scalpel, text marker and sterile microcentrifuge tubes were prepared and put under laminar flow cabinet. Frozen larvae were transferred to a laminar flow and thawed. Scalpel and tweezers were sterilized by flame. The larvae were transferred into the first petri plate filled with ethanol and cleaned. One by one they were transferred into another petri plate, and with a sterile scalpel, a few millimeters of the anterior part of the larvae were cut off. With sterile tweezers, the larval gut was pulled out and transferred into sterile microcentrifuge tube. Scalpel and tweezers were sterilized again and the process was repeated. Guts from the same box and the same sampling point were pooled in one microcentrifuge tube (at least 0.05 g gut/microcentrifuge tube). After gut extraction, microcentrifuge tubes were freeze until further use (Kannan et al., 2015).

3.2.6 DNA isolation, quantification and qualification

DNA was isolated from frozen gut samples using a NucleoSpin® Soil extraction kit. DNA extraction was done according to NucleoSpin® Soil User manual (2016) with a few modifications.

Before extraction, extra beads (one big glass (2r=1.0mm), three small silicon (2r=0.5mm)) were added to each NucleoSpin® Bead Tube Type A and autoclaved.

0.05 g of gut sample were weighted into pre-prepared NucleoSpin® Bead Tubes Type A. 700 µL Buffer SL1 and 150 µL Enchancer SX were added to the tubes. The tubes were put at 97°C at 400 rpm for 20 minutes. After incubation at high temperature, samples were mixed using vortex at high speed for 5 min. In the next step, tubes were centrifuged at 11000 x g for 2 min. After centrifugation, 150 µL of Buffer SL3 were added and tubes were mixed using vortex for 5 s. The samples were incubated at 4°C for 5 min and then centrifuged again for 1 min at 11000 x g.

All the following steps for binding and washing extracted DNA samples were done in the same way as written in the NucleoSpin® Soil User manual (2016). DNA was eluted twice with 30 µL PCR dH_2O.

Extracted DNA was stored at 4 °C until detection analysis.

Agarose gel electrophoresis was used to check if DNA extraction was successful. Electrophoresis was performed in TAE buffer. Agarose gel was prepared with GelGreen Nucleic Acid Strain, 10000x DMSO and 1% agarose. Gene Ruler Express DNA Ladder (0.5 μg/μL) was used as DNA ladder and a voltage of 100 V was chosen for the electrophoresis run.

Concentration of extracted DNA was checked using NanoDrop® measurement.

3.2.7 Sequencing

Extracted DNA samples were sent to Microsynth AG, Schützenstrasse 15, 9436 Balgach, Switzerland and sequenced with Illumina (plates) method using 515f and 806r (GTGCCAGCMGCCGCGGTAA and GGACTACHVGGGTWTCTAAT) primers of V4 16S rRNA (Klindworth et al., 2012).

3.2.8 Bioinformatical and statistical analysis

Analysis of the V4 16S rRNA sequences were performed with Mothur v.1.40.3. Sequences were trimmed based on index presence with less than two errors on primers; size comprised between 250 bp and 350 bp; and no ambiguous sequences and less than 8 homopolymers (polymer of identical monomer units - nucleotides). Sequences were then aligned on Silva v.132 and chimers (sequences formed from two or more biological sequences joined together) were removed with UCHIME implemented in Mothur (Schloss et al., 2009). For clustering, 97% similarity was set. Sequencing data

was subsampled considering the smallest sample and with the removing of the OTUs which had less than 2 (nseq=2) or 5 (nseq=5) members according to the all samples. Core microbiome was defined based on the presence of OTU in at least 80% of all samples. Data was analyzed using nonmetric multidimensional scaling (nMDS), analysis of molecular variance (AMOVA), homogeneity of molecular variance (HOMOVA) and one-way permutational multivariate analysis of variance (PERMANOVA) approaches. Similarity in community membership was calculated with jclass and similarity in community structure with Bray-Curtis and thetayc (Table 7). P-value smaller than 0.05 corrected for multiple comparisons was used for conformation of significant differences between samples as described before (Šket et al., 2017; Mothur, 2008).

Table 7: Similarity indexes used for statistical analysis.

Similarity index	Calculators	Definition
Jclass (Jaccard index)	Similarity in community membership	Is a statistic used for comparing the similarity and diversity of sample sets (Kosub, 2016).
Thetayc (Yue & Clayton theta similarity coefficient)	Similarity in community structure	Is a similarity index which includes the species proportions of both the shared and non-shared species in each population, and also propose a nonparametric maximum likelihood estimator for this index (Yue and Clayton, 2005)
Bray-Curtis	Similarity in community structure	Is a statistic used to quantify the compositional dissimilarity between two different sites, based on counts at each site (Bray and Curtis, 1957)

Multidimensional scaling (MDS) is a set of data analysis methods, which allow one to infer the dimensions of the perceptual spaces of subjects. The raw data entering into an MDS analysis are typically a measure of the global similarity or dissimilarity of the stimuli or objects under investigation. The primary outcome of an MDS analysis is a spatial configuration, in which the objects are represented as points. The points in this spatial representation are arranged in such a way, that their distances correspond to the similarities of the objects: similar object are represented by points that are close to each other, dissimilar objects by points that are far apart (Wickelmaier, 2003). There are two most used MDS analysis: classical and nonmetric. In nonmetric MDS, only the ordinal information in the proximities is used for constructing the spatial configuration (Wickelmaier, 2003).

Analysis of molecular variance (AMOVA) is a nonparametric analog of traditional analysis of variance. This method is widely used in population genetics to test the hypothesis that genetic diversity within two populations is not significantly different

from that which would result from pooling the two populations. The AMOVA statistic was calculated by

$$SS_{\mathrm{W}} = \frac{1}{n} \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} d_{ij}^2 \varepsilon_{ij} \qquad \ldots(8)$$

where n is the number of the sequences per treatment, N is the number of sequences, d_{ij} is the distance between sequences i and j, and ε_{ij} is 1 when i and j are from the same tretamnet and 0 when i and j are from the different treatments. A P-value is calculated by measuring the fraction of 1000 randomizations of the rows and columns in a distance matrix where the observed SSw is less than or equal to the randomized SSw values. Those P-values less than 0.05 were considered significant (Schloss, 2008).

Homogeneity of molecular variance (HOMOVA) is a nonparametric analog of Bartlett's test for homogeneity of variances, which has been used in population genetics to test the hypothesis that the genetic diversity within two or more populations is homogeneous. The HOMOVA statistic is calculated by

$$B = \frac{(N-P)\ln\left(\frac{SS_W}{N-P}\right) - \sum_{i=1}^{P}(N_i - 1)\ln\left(\frac{SS_{Wi}}{N_i - 1}\right)}{1 + \frac{1}{3(P-1)}\left(\sum_{i=1}^{P}\frac{1}{N_i - 1} - \frac{1}{N-P}\right)} \qquad \ldots(9)$$

where N is the total number of sequences in the study, P is the number of the treatments, N_i is the number of sequences in treatment i, and SSw_i is the amount of SSw contributed by treatment i. The P-value of the observed B is determined my measuring the fraction of 1000 randomizations of the rows and columns in the distance matrix, where the observed B is greater or equal to the randomized B-value. Those P-values less than 0.05 were considered significant (Schloss, 2008).

Permutational multivariate analysis of variance (PERMANOVA) is a geometric partitioning of multivariate variation in the space of a chosen dissimilarity measure according to a given ANOVA design, with p-values obtained using appropriate distribution-free permutation techniques (Anderson, 2017).

For the building of the dendrogram of the substrates OTUs table, Neighbor joining method with Bray-Curtis dissimilarity index was used.

For statistical analysis, PAST (PAlentological STatistics) Version 3.16 was used. Substrates and residuals data was analyzed using summary statistics and principal components analysis with correlation matrix and sequencing data was analyzed using one-way PERMANOVA approaches (Filipić, 2016).

4 RESULTS

4.1 CHARACTERISTICS OF SUBSTRATES AND RESIDUALS

All substrates used in the experiment were acidic (pH 4-6). Chicken feed was feed with the highest dry matter (90.13%), while oil residuals had the highest volatile solids content (99.79%). Chicken feed was the substrate with the highest chemical oxygen demand, but with the lowest ammonium content. The highest content of carbon was presented in oil residuals. Nitrogen content was higher in food residuals and chicken feed. Proteins were detected in all three substrates, with the highest concentration in chicken feed (3.588 mg/mL). With PAH BAH method, the highest amount of reducing sugar was detected in food residuals, while concentration of reducing sugar was under detection limit in oil residuals. On the other hand, oil residuals had the highest percent of fat. Volatile fatty acid profile was different between the substrates. Chicken feed contained high concentration of acetic acid and lactic acid in comparison to other two substrates. Oil residual was very rich in propionic acid relative to other two substrates. In comparison with food residuals and chicken feed, in oil residuals also isobutyric, isovaleric and valeric acid were detected. All the results are presence in Annex C.

At the end of feeding experiment residuals were taken out of the experiment boxes and measured for the same characteristics as substrates. Residuals showed higher active pH according to the substrates (box5-8 feed with oil residuals, box9-12 feed with food residuals), only exception was box 1-4 feed with chicken feed. Chicken feed residuals (box1-4) had the highest dry matter content, while oil residuals residues (box5-8) had the highest volatile solids content. We obtained the same results also in the substrates. In comparison with the substrate, ammonium amount increased over the experiment. In box 1-4 ammonium content was 20-times higher, in box 5-8 was 3-times higher and in box 9-12 the NH_4-N was more then 30-times higher. There were low increasing of total C and N in the experiment period. Protein concentration was higher in the residuals than in the substrates used. Reason may be present of bacterial protein in the residuals. There was no significant decreasing of reducing sugar detected in residuals comparing to the substrates. Fat content increased in boxes fed with food residuals and oil residuals. For the chicken feed, fat content did not increase significantly. Volatile fatty acid profile changed over the time and residuals contained different amount of VFA as the substrates. In comparison with the substrates, following was observed: (i) there was no formic acid detected in all three different residuals, (ii) concentration of acetic acid increased in box 9-12 and decreased in box 1-4 and 5-8, (iii) propionic acid increased in chicken feed boxes and decreased in all other boxes, (iv) butyric acid decreased in box 1-4 and 5-8, and increased in box 9-12, (v) lactic acid decreased in all the feeding residues, (vi) isobutyric and isovaleric acid increased in all the residues, and (vii) valeric

acid increased in box 5-8 and 9-12, but was still not detected in boxes fed with chicken feed. All the results are presence in Annex D.

On the Figure 3 PCA plot is presence. With PC1, 39.07% and with PC2, 23.357% of variance was explained. All substrates and residuals made separate groups. Residues from boxes feed with food residuals were less homogenous in comparison to other. COD, formic and lactic acids were the parameters which mostly effect on separation of chicken feed from other substrates and residuals.

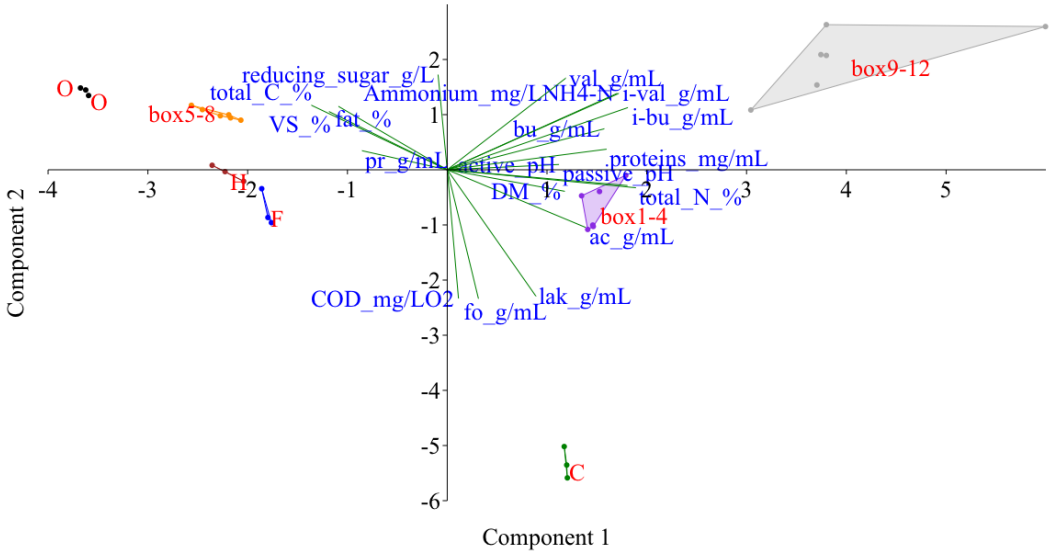

Figure 3: PCA (correlation) plot of substrate characteristics (oil residuals-O, food residuals-F, chicken feed-C, pine humus-H) and residuals (box1-4 – 1-4, box5-8 – 5-8, box9-12 – 9-12).

4.2 GROWTH OF BSF AND WASTE BIOCONVERSION

Larvae growth in box1-4 fed with chicken feed presented reference. They grew linearly with the highest weight at day 16 (day 14 according to weight/DM and weight/OM). Over the time, dry matter and organic matter were increasing. At the day 12, first prepupae were found. DM and OM of prepupae were almost the same as the DM and OM of the remaining larvae (day 14 and day 18).

Larvae in box 5-8 were fed with oil residuals. Growth was slow. After day 6 larvae started dying and no prepupae developed. After 12 days experiment was ended because of larvae death. On day 12, weight of the larvae was comparable to weight of the larvae fed with chicken feed at day 2 (Figure 4). DM and OM of the larvae at day 12 were higher as the DM an OM of the larvae feed with chicken feed at day 12.

In box 9-12, larvae were fed with food residuals. Growth was linear and fastest as in box 1-4. The highest weight of larvae was detected at day 14. OM and DM of the larvae were increasing over the time. At day 14, the first prepupae were developed. There OM and DM were almost the same as the DM and OM at the larvae last days (day 20). In comparison with the larvae fed with chicken feed, prepupae formation started later, but the organic matter of prepupae fed with food waste was much higher (95.96%) as the OM of the control (87.65%). The results are presence in Annex E and F.

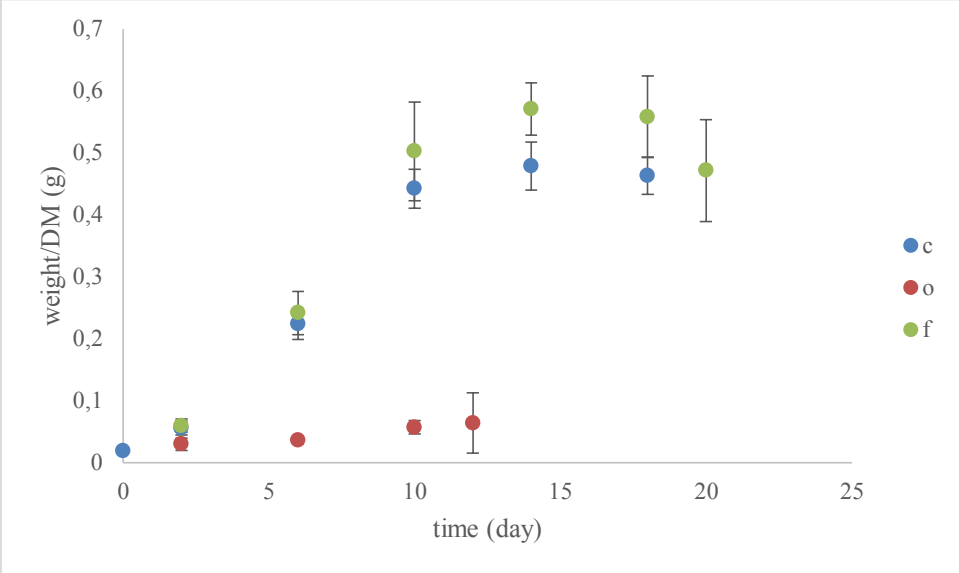

Figure 4: Weight of larvae according to DM (c-chicken feed, o-oil waste, f-food waste).

Biomass conversion rate was the highest in boxes 9 to 12 (larvae fed with food residuals) – 56.17%. Control biomass conversion rate was almost 10% lower (Table 8). In box5-8 (larvae feed with oil residuals) were no prepupae formation and BCR was not calculated.

Material reduction was the highest in control boxes (1-4, larvae fed with chicken feed), 70.18%. MR of larvae fed with food waste was much smaller, for almost 23% (Table 8). There was no material reduction ratio calculated for larvae fed with oil residuals, while results showed, organic matter of output material were higher than VS of the input material, which gave us negative material reduction ratio.

When putting MR and BCR data together: larvae growing on chicken feed was very good at material reduction (high MR), but reduced material was not convert into larvae biomass (low BCR) but went into the energy metabolism of the larvae or was lost other ways (bacteria).

Waste reduction index (equation (6)) was the highest in larvae boxes fed with chicken feed. For 1.5g/day lower value were detected in larvae fed with food waste. Overall

degradation was also the highest in boxes 1-4 (Table 8). Input material was (i) decomposed (metabolism), (ii) used for growth (larvae+pupae) or (iii) was not used by larvae (residues with microbial community). In the Table 8 balancing results are present. Larvae fed with chicken feed gave the best decomposing ratio, while larvae fed with food waste were best at larvae and pupae production.

Table 8: BCR, MR, WRI, D and balancing data of the feeding experiment (n=4).

	BCR (%)	MR (%)	WRI (g/day)	D (-)	Balancing		
					Material (%)	Larva and pupae (%)	Metabolism (%)
Box 1-4	47.82±0.83	70.18±0.44	3.82±0.03	0.69±0.005	29.79±0.43	3.92±0.11	66.29±0.38
Box 5-8	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A
Box 9-12	56.17±0.78	48.61±3.71	2.21±0.30	0.43±0.04	51.28±3.70	10.89±0.83	37.83±2.97

N.A. - not available

4.3 GUT MICROBIAL COMMUNITY

4.3.1 Substrates

Bacterial diversity between the different substrates on phylum level is presented in Figure 5. In the Table 9, number of the sequences of the substrate samples after trimming, filtration and subsample procedure are present. Around 50% of OTUs in chicken feed belong to phylum *Proteobacteria*, around 25% to *Firmicutes* and around 15% to *Bacteroidetes*. Dominant OTUs in food waste samples were members of *Firmicutes* (over 90%). Oil waste samples possessed more then 50% of phylum *Firmicutes* and around 40% of *Bacteroidetes*.

Table 9: Number of the bacteria and archaea sequences presented in the substrate samples after length trimming, homopolymers and chimers removing and number of the bacterial sequences after subsampling of the data (subsampled, nseq=2, nseq=5).

Substrate	Number of bacterial sequences	Number of archaea sequences	Subsampled samples (number of OTUs=459)	Subsampled samples; nseq=2 (number of OTUs=157)	Subsampled samples; nseq=5 (number of OTUs=102)
C1	1236	9	843	743	706
C2	1169	15	843	737	688
C3	843	19	843	741	680
O1	92039	82	843	831	815
O2	71629	73	843	817	800
O3	75201	80	843	827	806
F1	24752	8	843	829	823
F2	40593	14	843	835	835
Total	307462	300	-	-	-

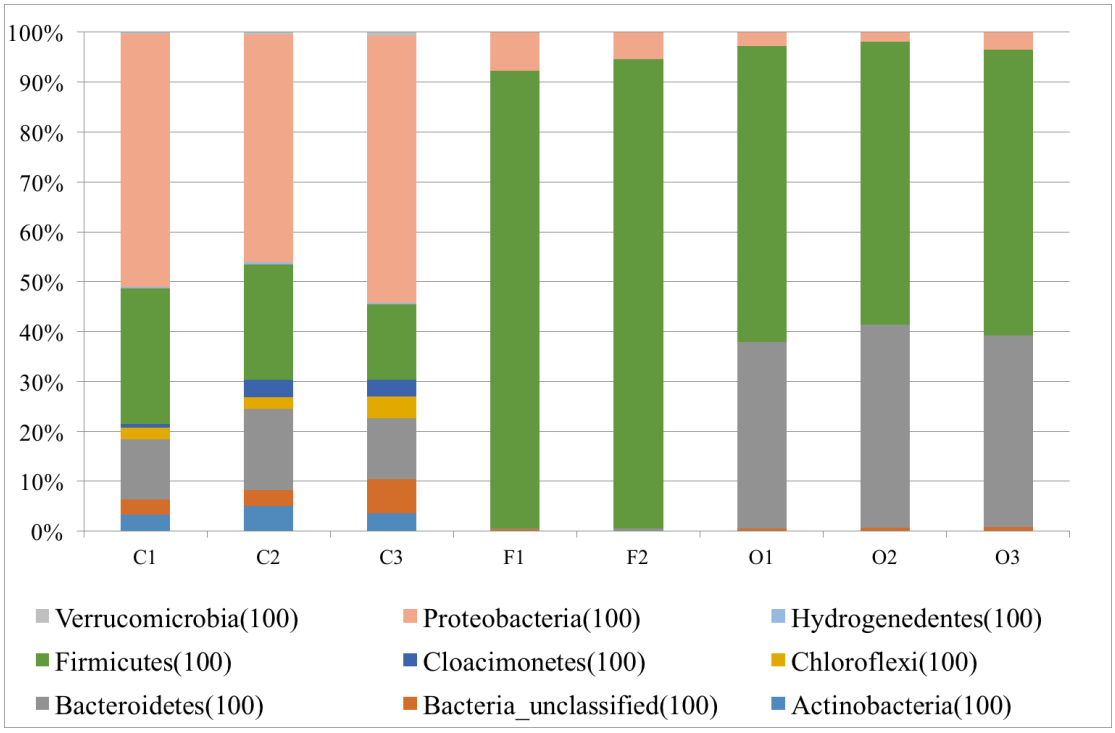

Figure 5: Bacterial OTUs presence in substrates - phylum level, nseqs=5, C-chicken feed, F-food waste, O-oil waste.

Dendrogram on the Figure 6 presents subsamples bacterial OTU data of the substrates. Oil waste and food waste are clustered together. OTUs of one of the chicken feed replicate is separated from the other two.

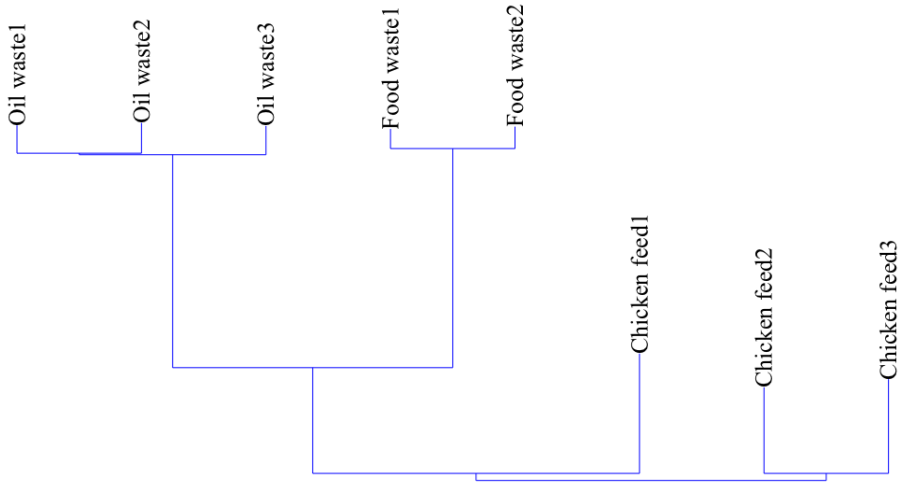

Figure 6: Dendrogram of the substrates according to Neighbor joining clustering and Bray-Curtis similarity index.

OTU data was tested with three different statistical tests, to determine significant differences between the input substrates (Annex G). AMOVA results with Thetayc

similarity index showed only chicken feed and oil residuals was significantly different between each other. Overall results showed substrates are not significantly different between each other.

4.3.2 Larvae gut community

Table 10 showing the number of the sequences in the larvae gut samples. After the trimming, filtering, aligning and chimers removing 2.769,644 bacterial sequences were present in the gut samples.

Table 10: Number of the sequences in the gut samples over the bioinformatical analysis.

Sample*	Input number of the sequences	Number of the sequences before chimer removing	Number of the sequences after chimer removing	Number of the sequences after aligning on Silva v.132	Number of the bacterial sequences	Number of the archaea sequences
Total number of the sequences	2800855	2800284	2772596	2772596	2769644	374
C0.1	38452	-	-	37759	37742	6
C0.2	83245	-	-	81544	81456	16
C0.3	92234	-	-	90588	90579	6
C2.1	87119	-	-	85771	85723	6
C2.2	87479	-	-	87327	87268	9
C2.3	80057	-	-	79768	79560	11
C10.1	78042	-	-	76770	76675	10
C10.2	81711	-	-	80529	80404	9
C10.3	80979	-	-	79431	79417	3
C14.1	62126	-	-	61775	61688	14
C14.2	57856	-	-	57615	57600	11
C14.3	73064	-	-	72398	72384	8
C18.1	62555	-	-	61754	61744	5
C18.2	69967	-	-	68976	68819	18
C18.3	55538	-	-	54532	54386	14
F2.1	101548	-	-	100838	100721	20
F2.2	65299	-	-	65222	65176	14
F2.3	114805	-	-	114363	114340	8
F10.1	93319	-	-	91773	91731	6
F10.2	101251	-	-	99942	99731	18
F10.3	81635	-	-	80408	80387	7
F14.1	101488	-	-	99428	99278	9
F14.2	98676	-	-	97510	97496	5
F14.3	37352	-	-	37306	36893	19
F20.1	88163	-	-	87614	87602	8
F20.2	95388	-	-	94526	94519	3

continued

continuation of Table 10.

Sample*	Input number of the sequences	Number of the sequences before chimer removing	Number of the sequences after chimer removing	Number of the sequences after aligning on Silva v.132	Number of the bacterial sequences	Number of the archaea sequences
F20.3	105680	-	-	104918	104840	18
O2.1	54422	-	-	54074	53935	9
O2.2	61476	-	-	61094	61028	11
O2.3	71842	-	-	71232	71109	15
O10.1	47134	-	-	47045	47034	7
O10.2	76338	-	-	75895	75876	7
O10.3	103741	-	-	103189	103168	8
O12.1	66634	-	-	66420	66270	16
O12.2	68639	-	-	67983	67878	10
O12.3	75601	-	-	75279	75187	10

*Letter stands for feeding condition; first number presenting day of sampling; the last number is the replicate.

OTU data (genus level) from larvae gut samples is presence in Annex H. There was high variability inside each of the sample.

OTU data was tested with the different statistical methods (AMOVA, HOMOVA, one-way PERMANOVA): (i) each of the sample with other sample (chicken feed gut sample at day 0 compared with chicken feed gut at day 20) and (ii) all samples with specific feeding condition with other samples with another feeding condition (all chicken feed gut samples compared with all food waste gut samples).

When testing each of the sample with other samples, no significant changes over time and between different substrates was detected (Table 11, Table 12). One-way PERMANOVA with Bray-Curtis similarity index showed no significant differences between larvae groups (per feed regiment/time of incubation) (Table 11). AMOVA method with thetayc index showed significant differences in the gut community of the larvae fed with food waste according to different age (F2 and F14; F2 and F20). Bacterial community at day 14 and day 20 was significantly different from community at day 2. Bacterial community of the larvae fed with food waste at day 2 was also significantly different from the gut microbiota of the larvae fed with chicken feed at day 10 and larvae fed with oil waste at day 12. Gut microbiota of the larvae fed with food waste at day 14 was significantly different from all the bacterial communities in the gut of the larvae fed with oil waste and from the larvae fed with chicken fed at day 14 (Table 12).

Table 11: One-way PERMANOVA results of subsampled gut OTU sequencing data. Method was performed with Bray-Curtis similarity index and 9999 number of permutation.

One-Way PERMANOVA, Bray-Curtis, permutation N=9999												
	C0	C10	C14	C18	C2	F10	F14	F2	F20	O10	O12	O2
C0		0.1043	0.1036	0.1014	0.102	0.1009	0.0995	0.104	0.0979	0.0983	0.1031	0.0919
C10	0.1043		0.6995	0.1983	0.102	0.7994	0.1031	0.0986	0.101	0.0978	0.1089	0.1043
C14	0.1036	0.6995		0.5987	0.0981	0.7024	0.1011	0.0984	0.0947	0.1021	0.0995	0.0967
C18	0.1014	0.1983	0.5987		0.1022	0.2049	0.1012	0.1061	0.0981	0.097	0.0937	0.1036
C2	0.102	0.102	0.0981	0.1022		0.0965	0.0997	0.2016	0.1006	0.1017	0.0953	0.0964
F10	0.1009	0.7994	0.7024	0.2049	0.0965		0.3015	0.2033	0.0978	0.0995	0.1012	0.1016
F14	0.0995	0.1031	0.1011	0.1012	0.0997	0.3015		0.0993	0.1008	0.0944	0.099	0.0976
F2	0.104	0.0986	0.0984	0.1061	0.2016	0.2033	0.0993		0.0976	0.0963	0.0982	0.1986
F20	0.0979	0.101	0.0947	0.0981	0.1006	0.0978	0.1008	0.0976		0.0997	0.1013	0.0988
O10	0.0983	0.0978	0.1021	0.097	0.1017	0.0995	0.0944	0.0963	0.0997		0.9008	0.1022
O12	0.1031	0.1089	0.0995	0.0937	0.0953	0.1012	0.099	0.0982	0.1013	0.9008		0.1024
O2	0.0919	0.1043	0.0967	0.1036	0.0964	0.1016	0.0976	0.1986	0.0988	0.1022	0.1024	

Table 12: AMOVA results of the subsamples gut OTU sequencing data. Method was performed with Thetayc similarity index.

AMOVA - thetayc												
	C0	C2	C10	C14	C18	F2	F10	F14	F20	O2	O10	O12
C0		0.310	0.098	0.208	0.323	0.143	0.094	0.056	0.109	0.090	0.108	0.094
C2	0.310		0.098	0.101	0.095	0.250	0.101	0.054	0.101	0.125	0.104	0.094
C10	0.098	0.098		0.061	0.305	0.044	0.512	0.061	0.111	0.110	0.098	0.102
C14	0.208	0.101	0.061		0.716	0.298	0.813	0.045	0.084	0.092	0.109	0.088
C18	0.323	0.095	0.305	0.716		0.271	0.576	0.110	0.110	0.096	0.096	0.091
F2	0.143	0.250	0.044	0.298	0.271		0.185	0.023	0.049	0.242	0.057	0.038
F10	0.094	0.101	0.512	0.813	0.576	0.185		0.218	0.105	0.100	0.207	0.092
F14	0.056	0.054	0.061	0.045	0.110	0.023	0.218		0.061	0.036	0.049	0.047
F20	0.109	0.101	0.111	0.084	0.110	0.049	0.105	0.061		0.106	0.103	0.120
O2	0.090	0.125	0.110	0.092	0.096	0.242	0.100	0.036	0.106		0.113	0.108
O10	0.108	0.104	0.098	0.109	0.096	0.057	0.207	0.049	0.103	0.113		1.000
O12	0.094	0.094	0.102	0.088	0.091	0.038	0.092	0.047	0.120	0.108	1.000	

Non-metric multidimensional scaling (each of the sample with other samples) produced acceptable stress value 0.1419 and R^2 0.8263 (axis1: 0.6215, axis2: 0.1948). Gut community of the larvae fed with chicken feed is separated from the other communities. Microbiota of the larvae fed with oil waste and food waste was changing over the time (Figure 7).

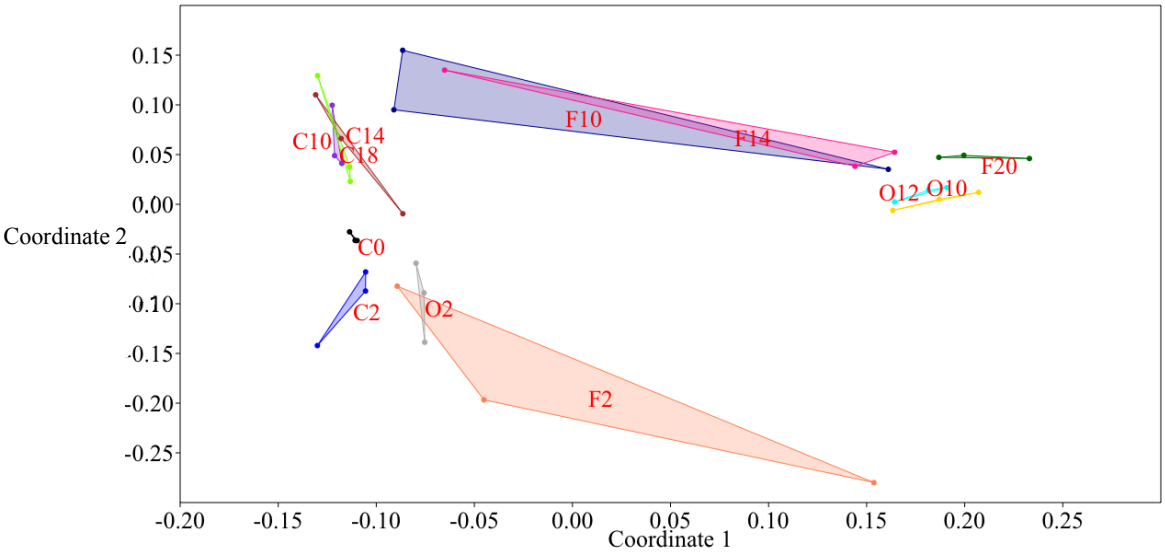

Figure 7: Non-metric multidimensional scaling plot with Bray-Curtis similarity index of subsampled larvae gut OTU data (letter stands for feeding condition; number stands for day of sampling).

When comparing all chicken feed, food waste and oil waste guts together, sequencing data was significantly different between chicken feed guts and food waste guts and between chicken feed guts and oil waste guts (Table 13).

Table 13: One-way PERMANOVA results of subsampled gut OTU sequencing data (all data from one feeding condition together as one sample). Method was performed with Bray-Curtis similarity index and 9999 number of permutation.

	Chicken feed guts	Food waste guts	Oil residuals guts
Chicken feed guts		0,0001	0,0001
Food waste guts	0,0001		0,1813
Oil residuals guts	0,0001	0,1813	

nMDS (all samples with specific feeding condition with other samples with another feeding condition) produced acceptable stress value 0.1419 and R^2 0.8263 (axis1: 0.6215, axis2: 0.1948). Oil waste OTU data and food waste data are overlapping. Food waste OTU data is very variable, his surface is much bigger from the other two feeding condition data. Oil waste OTU data is part of the food waste data. Chicken feed OTU data is almost separated from the food and oil waste data. Exception is one of the replicates at day 14 (C14.1) (Figure 8).

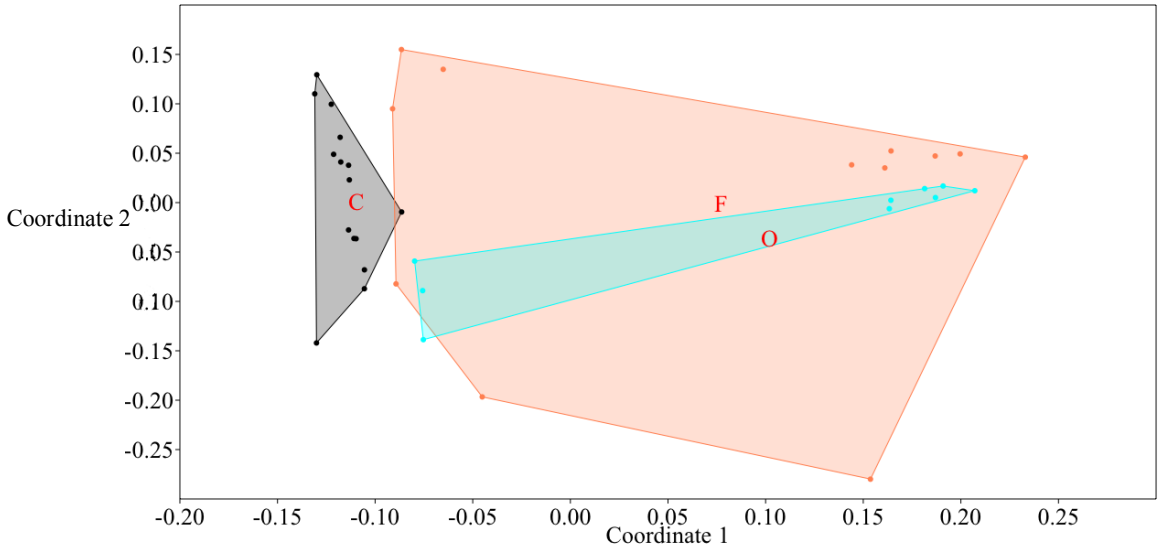

Figure 8: Non-metric multidimensional scaling plot with Bray-Curtis similarity index of subsampled larvae gut OTU data (C-larvae fed with chicken feed; F-larvae fed with food waste; O-larvae fed with oil waste).

The most dominant Bacterial phylum presented in larvae gut samples were *Proteobacteria*, *Firmicutes* and *Bacteriodetes*.

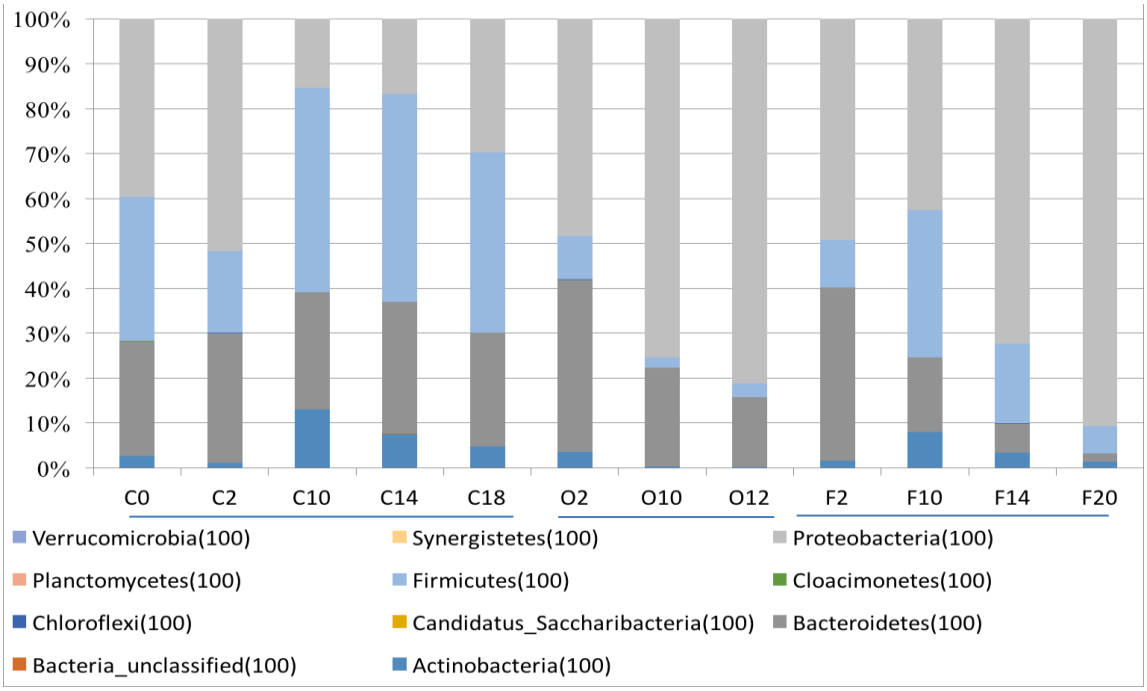

Figure 9: Bacterial phylum in larvae gut samples (97% similarity approach, nseqs=5). First five columns from left to right presenting microbial community in larvae guts fed with chicken feed from day 0 to day 18 (C0, C2, C10, C14, C18). The following three columns presenting guts of larvae fed with oil residuals from day 2 to day 12 (O2, O10, O12) and the last four columns presenting data for larvae fed with food waste from day 2 to day 20 (F2, F10, F14, F20).

4.3.3 Core microbiome

Core microbiome of the gut samples were determinate based on two different approaches (97% similarity approach and phylotypes on genus level). Core was defined for Bacteria and Archaea separately. Criteria for the core was present of the sequence in 80% of all the samples.

Bacterial core microbiome is constructed of *Enterococcus, Enterobacteriacea_* unclassified and *Dysgonomonas*. Those three OTUs presented almost 48% (based on 97% similarity approach) of all the sequences in the samples (Table 13).

Archaea core microbiome contained *Methanobacterium* and *Methanothrix*. This two genus presented 58% (based on 97% similarity approach) of all the sequences in the samples (Table 14).

Table 14: Core microbiome of gut samples.

Bacteria		Archaea	
97% similarity	Phylotype	97% similarity	Phylotype
Enterococcus *Enterobacteriaceae_* unclassified *Dysgonomonas*	*Enterobacteriaceae_* unclassified *Dysgonomonas* *Enterococcus*	*Methanobacterium* *Methanothrix*	*Methanobacterium* *Methanothrix*
47.97% of all sequences	**54.42%** of all sequences	**58.02%** of all sequences	**63.90%** of all sequences

5 DISCUSSION

Substrates used in the experiment differed in physical and chemical parameters. Over the experiment period, substrates were used by larvae. Substrates were degraded, reduced and used for living needs and reproduction of larvae.

Residuals, collected at the end of experiment period, contained less organic matter according to the initial substrate because larvae used them. Ammonium content in substrates was very low, but in residuals its content increased. This may be resulting on substrate decomposition and feces produced by larvae, and could represent a valuable additive for composting and as a useful additive in agriculture for plotting flowers.

Protein and fat content was higher in residuals. Reason may be in production of fatty and protein rich feces by larvae. On the other hand, concentration of reducing sugar was lower in residues. Monosaccharides and disaccharides are easy to degrade and present good source of energy. Protein and fat rich residuals collected at the end of feeding experiment could be used as a soil amendment and could present good nutrition source for plants. According to short processing time, the residue needs to undergo a maturation phase in order to prevent oxygen depletion in the soil which inhibits seed germination or suppresses root and plant growth (Brinton and Evans, 2001).

Volatile fatty acid profile changed in the time of incubation. In all three feeding conditions iso-butyric and iso-valaric acid concentration increased. In the OTU data *Clostridium* and *Megasphera* were presented. Those two genera are known bio-butyric acid producers (Jha et al., 2014).

Oil residuals were not suitable substrate for larvae growth. After 12 days of feeding with this substrate, all of the larvae died. Their weight was increasing very slowly and their weight was significantly different from larvae fed with food waste and chicken feed, from the first sampling point (day 2). Substrate full of fat in organic matter resulted in very high organic content in those larvae. According to Nguyen et al. (2015), similar results were observed. Fish diet had high fat and calorie content, but was consumed at a slower rate than kitchen waste. They suggested, substrate contain component which sow down the larvae growth. In his case this were heavy metals in the fish diet. In my experiment, presence of some cleaning fluid used for dish washing, could affect negatively on larvae growth on oil residuals. Another possibility for slow growth and potentially death of the larvae fed with oil waste, may be substrate preparation. Substrate was collected from the reservoir. In it, water, oil and the rest of wastes were separated from each other. Maybe feeding with just specific layer without mixing of all the reservoir content, would give us better growth.

Food waste and chicken feed were suitable substrates for larvae growth, prepupae formation and waste reduction. Their weight was increasing linearly in first three sampling points (day 2 – day 10). At the incubation day 14, weight of larvae fed with food waste was significantly higher than weight of larvae fed with control substrate. The same difference was observed in another sampling point (day 18). Different composition of the substrates (food waste, chicken feed) affected on the organic matter of the larvae. Organic content of larvae fed with food waste raised from initial 83% to 93% already in the first sampling point. Increasing of OM in larvae fed with chicken feed was slower and reached only 87.4% at the last sampling day (day 18). Larvae composition was affected by the substrate and this has significant impact on downstream usability of larvae as feedstuff for other animals (Ruhnke et al., 2018).

Prepupae formation was first detected in boxes 1-4, at day 12 (age of larvae 18-19 days). In boxes 9-12, first prepupae were detected at day 14 (age of larvae 20-21 days). Weight of prepupae was significantly higher in boxes 9-12 and their organic matter content was also higher in those boxes. In comparison with the Newton et al. (2005), ash (an-organic) content of the dry prepupae in his experiment was much higher (14.6% for BSF fed with poultry and 16.6% for BSF fed with swine manure) than an-organic content of prepupae formed on chicken feed (12.4%) and food wastes (4.0%).

According to Smet et al. (2018), larvae stage least from 13 to 18 days. In my feeding experiment larvae stage lasted longer, in boxes 1-4 from 18 to 25 days and in boxes 9-12 from 20 to 27 days. Optimal growth conditions were not reached in this feeding experiment. Reason might be in the substrates used.

Biomass conversion rate was higher in boxes 9-12. On the other hand, material reduction was better in boxes 1-4, for 21%. Larvae fed with chicken feed transformed 70% of input organic material into volatile compounds, such as CO_2 or NH_3, or stored it as biomass in the adult larvae that were removed from the system as prepupae (Dortmans, 2015).

Waste reduction index was higher for the larvae fed with chicken feed (3.8 g/day) and was as same as Diener et al. (2009) reported for optimal feeding rates.

Control conditions (larvae fed with chicken feed) used more than two third of the input material for metabolism and larvae and pupae production. In comparison, larvae fed with food waste used less then 50%. Residue/material that was not used for larvae development and transformed into volatile compounds by larvae, can be used as a soil amendment (Riuji Lohri et al., 2017). According to this, larvae fed with chicken feed were better metabolic convertor of the input material, but gave less material for used as a soil amendment. Larvae fed with food waste needed more material for larvae and

pupae development and also gave more material which could be further used in agriculture. Those results showed different substrates gave different benefits. For production of heavier, organic richer pupae and bigger amount of material used in agriculture, food waste present better substrate. For higher metabolism of the input material and higher material reduction, chicken feed is better.

Increasing further the energetic content of the oily feedstuff did not prove to be efficient means of further increasing the quality of larvae. Hence additional approaches testing for introduction of more lignocellulose substrates as co-substrates would probably help increase the use of oily substrates in rearing of larvae.

Microbial community in the gut samples was different from the community in the input material. Control sample – larvae gut community at the beginning of the feeding experiment (C0) – was fed with chicken feed from the egg hatching day till the beginning of the experiment. Initial community of the substrate (~ 50% of Proteobacteria, ~ 25% of *Firmicutes*, ~ 15% of *Bacteroidetes*) changed in the gut of larvae. Presence of *Proteobacteria* OTUs decreased to ~ 40%. On the other hand, *Firmicutes* (~ 32%) and *Bacteroidetes* (~ 25%) presence increased. Over the life time, another phylum increased, *Actinobacteria*, but initial three phyla still stayed dominant.

Initial community of oil waste samples contained ~ 40% of *Bacteroidetes* and over 50% of *Firmicutes*. After two days of larvae incubation on this substrate, abound 75% of the Bacterial gut microbiota presented *Proteobacteria*. Percent of *Bacteroidetes* decreased to 20% and was still decreasing over the time. *Firmicutes* presents was decreasing too. After two days, there was just ~ 3% of their OTUs present in the community.

Food waste samples contained high percent of *Firmicutes* (more then 90%). After two days of larvae incubation on food substrate, present of *Firmicutes* OTUs decreased to around 10%. On the other hand, presence of *Proteobacteria* was increasing over the time, from 50% of population at day 2, to 90% of community at day 20. In the initial substrate, low presence of *Bacteroidetes* was detected. In the larvae gut, their population increased at day 2 to ~ 38%, but over the incubation time their presence still decreased.

In contrast, previous studies obtained different proportions of the same taxa. Jeon et al. (2011) reported, gut community of the larvae fed with food waste contained 67.36% of *Bacteriodetes*, 18.85% of *Proteobacteria*, 9.4% of *Firmicutes*, 2.01% of *Fusobacteria* and 1.92% of *Actinobacteria*. Experiment design between Jeon et at. (2011) and my feeding experiment was different. They started with different diets at the time of egg hatching. In comparison, my larvae were first few days fed with chicken feed and after that the diet was changed to food waste or oil residuals. On the other hand, they did not

analyze substrate community and there is no data how substrate community changed in the larvae gut.

Till now, there were no gut community study done with the substrates similar to oil waste substrate.

Overall, larvae gut community was composed by *Proteobactera*, *Firmicutes* and *Bacteriodetes*. In some samples some OTUs of *Actinobacteria* were detected. Substrate community had no noticeable affect on changes in larvae gut microbiota when comparing OTU data on phylum level of each of the sample with other samples.

There were some significant differences ($p \leq 0.05$) between gut community samples over the life time of larvae fed with food wastes (F2 and F14; F2 and F20). Gut community at the beginning of the feeding with new substrate changed over the continuous presence of this substrate.

Bacterial community showed some significant differences between different substrates, but overall results did not showed significance between gut community in larvae fed with chicken feed, oil waste and food waste. There is a chance, that larvae gut community would be significantly different between different diets, if the larvae were fed with specific substrate from the egg hatching time. Maybe core gut community was maintained in the first days, when larvae were fed with chicken feed and their structure composition was so solid, that other species (substrate community) were not capable to grow over them. The starting community may produce some antimicrobial substances, which prevented growth of adding community.

Cohousing of affected and unaffected (the same animals with the different microbial community) animals already harboring complex microbial population presenting one of the simplest methods of assessing the influence of a complex gut microbiota on a recognized phenotype (Ericsson and Franklin, 2015). All 2400 larvae used in the experiment were cultivated together and fed with the same substrate. According to the community results, this time may be crucial for gut community formation. In the time of incubation of the whole larvae community together (from egg hatching and before feeding experiment) presence of cohousing effect may be detected.

Gut microbiota differ greatly amongst laboratory animals, and these differences result in notable differences in experimental results. For example, mice of the same strain from different vendors have different microbiota profiles and similarly the same mice housing at different institutions have different microbiota profiles (Pluznick, 2014). This institution effect may be sensed also in my experiment and might be reason for different results comparing to other researches.

6 CONCLUSION

With this thesis BSF larvae growth on three different biowastes was tested. It was also checked if biowaste type has any affect on the growth rate and organic matter content of larvae and how the larvae bacterial gut microbiota is changing over the time and between different substrates.

Results showed, chicken feed is best degraded by larvae and give the highest waste reduction index. On the other hand, the highest yield in larvae biomass (more organic matter, bigger and heavier pupae) was observed on the food waste substrate.

The microbial community data collected from larvae intestinal tract indicated, substrate has no significant influence on the gut microbiota. Bacterial community was very variable. Three dominant domain was detected: *Proteobacteria*, *Firmucutes* and *Bacteroidetes*.

Core microbiome was detected and presented around 50% of all the sequences. According to high core and no significance differences between gut communities, specific species was not checked.

7 SUMMARY

7.1 SUMMARY

Larvae of the black soldier fly, *H. illucens* (Diptera: Stratomydiae), are voracious feeders of organic material and may be used in simple engineered systems to reduce organic waste in low- and middle-income countries (Diener et al., 2009).

In this master thesis, larvae of the black soldier fly were used to observe their growth on three different biowastes/substrates, chicken feed - used as a control, food waste – mixture of food residuals from Mensa at the UIBK and oil residuals – mixture of scrap oil, food residuals, already used water from washing machine and other residues from Mensa kitchen at the University of Innsbruck. Organic content of growing larvae, their development to prepupae stage and waste reduction were analyzed. With larvae sampling over the time, gut microbiota was isolated, sequenced and analyzed.

Results showed control substrate and food wastes are suitable substrates for larvae growth, prepupae development and waste reduction. On the other hand, oil waste was not appropriate substrate for the larvae. There were no prepupae development, waste reduction and after 12 days all of the larvae were death.

Food waste was the substrate that gave the heaviest larvae with high organic matter content. Chicken feed was the substrate on which prepupae formation begin first and gave the highest biowaste reduction index.

Sequencing data of the larvae gut microbiota was very variable. When comparing each of the sample with other samples, the community was not affected by the substrate microbiota and over the life period. When comparing all samples with specific feeding condition with other samples with another feeding condition, significant differences between chicken feed gut community and oil waste gut community, and between chicken feed gut community and food waste gut community was detected. Bacterial community was composed of three main phyla, *Proteobacteria*, *Firmicutes* and *Bacteroidetes*.

7.2 POVZETEK

Sodobna populacija se sooča s številnimi problemi, kot so naraščanje prebivalstva, večanje mest, večje potrebe po hrani in naraščanje odpadkov.

V zadnjem času se je oblikovala nova skupina odpadkov – mestni trdni odpadki (MSW-municipal solid waste) katere glavni proizvajalci so gospodinjstva, manjša podjetja in inštituti. Značilnosti teh odpadkov so visok delež organskih, biološko razgradljivih odpadkov (bio-odpadki) (Karak in sod., 2012; Riuji Lohri in sod., 2017).

Tehnologij za predelavo in zmanjšanje teh odpadkov je vedno več in so razdeljene v štiri kategorije: (i) direktna uporaba, (ii) biološka obdelava, (iii) fizikalno-kemijska obdelava in (iv) toplotno-kemična obdelava (Riuji Lohri in sod., 2017).

Ena izmed novejših in razvijajočih se tehnologij za biološki način obdelave bio-odpadkov predstavlja obdelava le teh z ličinkami črne vojaške muhe (BSF). Tu ličinke transformirajo odpadke v žuželčje proteine in maščobe (Riuji Lohri in sod., 2017). Ličinke črne vojaške muhe (black soldier fly – *H. illucens*) se lahko prehranjujejo z visokim naborom organskega materiala hkrati pa proizvedejo bube, ki se jih enostavno izloči iz združbe in predstavljajo dragoceno in hranljivo krmo za živali. Z ekstrakcijo maščob iz ličink in bub pridobimo primeren substrat za produkcijo biogoriv. Odpadki odpadkov, ki se ne pretvorijo v biomaso ličink in bub predstavljajo visoko hranljiv dodatek za gnojenje zemlje, hkrati pa predstavljajo substrat za produkcijo sladkorja (Nguyen in sod., 2015; Li in sod., 2011).

Namen magistrske naloge Uporaba črne vojaške muhe (*Hermetia illucens*) za produkcijo proteinov iz organskih odpadkov in analiza njenega mikrobioma je bilo (i) testiranje kako ličinke rastejo na treh različnih bio-odpadkih, (ii) ali ima substrat kakršen koli vpliv na hitrost rasti in vsebnost organskega materiala v ličinkah in (iii) kako se ličinkina bakterijska združba prebavil spreminja skozi čas in med različnimi substrati.

Pričakovano je bilo, da bodo rezultati razložili, kateri substrat je najbolj dovzeten za razgradnjo s strani ličink in na katerem substratu pride do produkcije največjih, najtežjih in organsko najbolj bogatih ličink/bub. Mikrobna združba prebavil ličinke bi pokazala, ali ima substrat/bio-odpadek vpliv na združbo prebavil, ali je jedrni mikrobiom prisoten in kakšen je, ter ali obstaja kakšna specifična vrsta bakterij, ki je prisotna le v prebavilih ličink, ki so bile hranjene z določenim substratom.

Za izvedbo eksperimenta hranjenja je bilo uporabljenih 2400 ličink črne vojaške muhe izleženih med 10.3. in 11.3.2017. Hranjene so bile s tremi različnimi substrati: (i)

kokošja krma, kot kontrolni substrat, (ii) ostanki hrane (zmiksana vsebnost smetnjaka za biološke odpadke) in (iii) ostanki olja (oljna plast in plast ostankov hrane iz rezervoarja, v katerem se zbira odpadno olje, voda iz pomivalnih strojev in ostanki hrane iz kuhinje na Univerzi v Innsbrucku). Dne 17.3.2017 se je eksperiment hranjenja pričel in je trajal 20 dni. Po 200 ličink je bilo razdeljenih v plastične posode. Po štiri posode so bile hranjene z istim substratom (škatla 1, 2, 3 in 4 z kokošjo krmo; škatla 5, 6, 7 in 8 z odpadnim oljem; škatla 9, 10, 11 in 12 z odpadki hrane). Hranjenje je bilo izvedeno vsak drugi dan. Prav tako, se je vsak sodi dan stehtalo posode in ličinke ter pobralo bube in mrtve ličinke, če so bile prisotne. Na vsak četrti dan je bilo odstranjenih 10 larv iz vsake posode. 5 se jih je uporabilo za določanje suhe in organske vsebnosti, preostalih pet pa se je shranilo za izolacijo prebavil in DNA mikrobne združbe GIT ličink. Po končanem eksperimentu smo shranili tudi ostanke prisotne v posodah, za nadaljnjo karakterizacijo.

Substrati in ostanki so bili ovrednoteni s pomočjo fizikalnih in kemijskih metod (pH, suha in organska snov, vsebnost amonija, kemijska potreba po kisiku, vsebnost ogljika in dušika, delež proteinov, maščob in redukcijskih sladkorjev, HPLC).

Iz substratov in prebavil ličink je bil izoliran dedni material, ovrednoten z gelsko elektroforezo in NanoDrop-om, ter poslan na sekvenciranje.

S pomočjo Mothur programa, je bila narejena bioinformacijska analiza, s pomočjo PAST programa pa statistična analiza dobljenih rezultatov.

Substrati so bili rahlo kisli. Kokošja krma je imela največji odstotek suhe snovi, najvišjo kemijsko potrebo po kisiku in največji delež proteinov. Odpadki olja so imeli največji odstotek organske snovi in ogljika, delež redukcijskih sladkorjev pa je bil pod mejo detekcije. Odpadki hrane so bili najbolj bogati v redukcijskih sladkorjih, v primerjavi z ostalima dvema substratoma. Profil hlapnih maščobnih kislin je bil različen med substrati.

Rezultati analize odpadkov, po koncu eksperimenta hranjenja, so pokazali, da je tekom eksperimenta delež amonija in proteinov v odpadkih narastel (v primerjavi s substrati) in da se je spremenil profil hlapnih maščobnih kislin.

S PCA analizo je bilo pokazano, da so substrati in ostanki lepo ločeni med seboj. Na ločitev kokošje hrane od ostalih vzorcev je najbolj vplival delež mravljične in mlečne kisline ter kemijska potreba po kisiku.

Rast ličink na kokošji hrani je bila linearna. Največjo maso so dosegle 16 dan eksperimenta. Organska in suha snov sta tekom poskusa naraščali. Prve bube so se

pojavile 12 dan eksperimenta in so bile po sestani (suha in organska snov) podobne ličinkam v zadnjih dneh pred prehodom v naslednji življenjski stadij.

Ličinke hranjene z odpadki olja so rasle zelo počasi. Po 6 dnevu poskusa so pričele umirati in 12 dan ni bilo več živih ličink. Imele so nizko maso, a visok delež suhe in organske snovi.

BSF rastoče na odpadkih hrane so rasle hitro in njihova masa se je linearno povečevala snovi čas. Suha in organska snov je bila višja kot pri ličinkah hranjenih s kokošjo krmo. Prve bube so se pojavile 14 dan poskusa in so bile po sestavi (suha in organska snov) podobne sestavi ličink v zadnjih dneh pred prehodom v naslednji stadij.

Vstopni materiali (kokošja krma, odpadki olja, odpadki hrane) so bili tekom poskusa razgrajeni in uporabljeni za rast in razvoj ličink.

Ličinke hranjene z odpadki hrane so največji delež vstopnega materiala porabile za rast (najvišji BCR – 56,17 %). Ličinke hranjene s kokošjo krmo pa so bile najbolj uspešne pri zmanjšanju vstopnega materiala (najvišji MR). Visok MR pri ličinkah hranjenih s kokošjo krmo se ujema z najvišjim indeksom zmanjšanja odpadkov (WRI).

Proteobacteria, *Firmicutes* in *Bacteroidetes* so bila najbolj zastopana debla v substratih. Kokošja krma je bila najbolj bogata s *Proteobacteria* (~ 50 %), ostanki hrane s *Firmicutes* (preko 90 %), populacija v odpadne olju pa sta predvsem zastopali debli *Firmicutes* in *Bacteroidetes*. Signifikantna različnost med substrati je bila testirana s tremi statističnimi metodami. AMOVA je pokazala, da sta odpadki olja in odpadki hrane signifikantno različna od kokošje krme, sama pa se med seboj ne razlikujeta signifikantno.

Mikrobna združba prebavil ličink znotraj posameznih vzorcev je bila zelo raznolika. S statističnimi metodami smo testirali ali je med vzorci prisotna signifikantna različnost. Ob testiranju vsakega vzorca z vsakim (One-way PERMANOVA, AMOVA, HOMOVA) nismo zaznali signifikantne različnosti med njimi. Rezultat nMDS analize so pokazali, da se združba v prebavilih ličink hranjenih s kokošjo krmo skozi čas ni signifikantno spreminjala, nasprotno pa je bilo opazno pri vzorcih ličink hranjenih z odpadnim oljem in odpadki hrane.

Ob obravnavanju vseh vzorcev mikrobne združbe hranjenih z istim substratom kot eno, smo statistične teste ponovili. Pokazala se je signifikanta razlika med združbo hranjeno s kokošjo krmo in ostanki hrane, in med združbo hranjeno s kokošjo krmo in odpadnim oljem. nMDS analiza je pokazala ločitev mikrobne združbe prisotne v prebavilih ličink hranjenih s kokošjo krmo od ostalih vzorcev. Vzorec mikrobne združbe prebavil ličink

hranjenih z odpadnim oljem pa je predstavljal podskupino znotraj vzorcev mikrobne združbe ličink hranjenih z odpadki hrane.

Najbolj zastopana debla v prebavilih ličink so bila *Proteobacteria*, *Firmicutes* in *Bacteriodetes*.

Bakterijski jedrni mikrobiom so sestavljale *Enterococcus, Enterobacteriacea_* unclassified in *Dysgonomonas*. Predstavljale so kar 48 % vseh sekvenc. Arhejski jedrni mikrobiom sta sestavljali *Methanobacterium* in *Methanothri.* Med vsemi arhejskimi sekvencami jih je bilo 58 %.

Eksperiment hranjena je pokazal da je kokošja krma substrat, ki ga ličinke črne vojaške muhe najuspešneje razgrajujejo (najvišji WRI). Nasprotno pa odpadki hrane predstavljajo hranilo, ki da največji izplen ličinkine biomase (najtežje in organsko najbolj bogate ličinke in bube).

Mikrobna združba prebavil je zelo raznolika znotraj vsakega vzorca. S pomočjo statističnih analiz nismo mogli potrditi signifikantno spreminjanje mikrobne združbe skozi čas in glede na različni vstopni substrat.

8 REFERENCES

Anderson M. J. 2017. Permutational multivariate analysis of variance (PERMANOVA). V: Wiley StatsRef: Statistics reference online. Balakrishnan N., Colton T., Everitt B., Piegorsch W., Ruggeri F., Teugels J. L. (eds.). doi: 10.1002/9781118445112.stat07841: 15 p.

Banks I. J., Gibson W. T., Cameron M. M. 2014. Growth rates of black soldier fly larvae fed on fresh human faeces and their implification for improving sanitation. Tropical Medicine and International Health, 19, 1: 14-22

Boccazzi I. V., Ottoboni M., Martin E., Comandatore F., Vallone L., Spranghers T., Eeckhout M., Mereghetti V., Pinotti L., Epis S. 2017. A survey of the microbiota associated with larvae of the black soldier fly (*Hermetia illucens*) reared for feed production. PLoS ONE, 12, 8: e0182533, doi: 10.1371/journal.pone.0182533: 15 p.

Bray J. R., Curtis J. T. 1957. An ordination of upland forest communities of southern Wisconsin. Ecological Monographs, 27, 4: 326-349

Brinton W. F., Evans E. 2001. How compost maturity affects container grown plants. Biocycle, 42, 1: 56-60

Diclaro II J.W., Kaufman P.E. 2009. Black soldier fly *Hermetia illucens* Linnaeus (Insecta: Diptera: Stratiomyidae). EENY 461. Gainesville, Entomology and Nematology Department, UF/IFAS Extension: 4 p. http://edis.ifas.ufl.edu/pdffiles/IN/IN83000.pdf

Diener S., Studt Solano N. M., Gutierrez F. R., Zurbrugg C., Tockner K. 2011. Biological treatment of municipal organic waste using black soldier fly larvae. Waste Biomass Valor, 2: 357-363

Diener S., Zurbrugg C., Tockner K. 2009. Conversion of organic material by black soldier fly larvae: establishing optimal feeding rates. Waste Management & Research, 27: 603-610

Diener S., Zurbrugg C., Tockner K. 2015. Bioaccumulation of heavy metals in the black soldier fly, *Hermetia illucens* and effects on its life cycle. Journal of Insects as Food and Feed, 1, 4: 261-270

Dortmans B. 2015. Valorisation of organic waste – effect of the feeding regime on process parametrs in a continuous black soldier fly larve composting. Master's thesis. Uppsala, Swedish University of Agricultural Sciences, Department of Energy and Technology: 38 p.

Elhag O. Zhou D., Song Q., Soomro A. A., Cai M., Zheng L., Yu Z., Zhang J. 2017. Screening, expression, purification and functional characterization of novel antimicrobial peptide genes from *Hermetia illucens* (L.). PLoS ONE, 12, 1: e0169582, doi: 10.10371/journal.pone.0169582: 15 p.

Ericsson A. C., Franklin C. L. 2015. Manipulating the gut microbiota: methods and challenges. ILAR Journal, 56, 2: 205-217

Erickson M. C., Islam M., Sheppard C., Liao J., Doyle M. P. 2004. Reduction of *Escherichia coli* O157:H7 and *Salmonella enterica* serovar Enteritidis in chicken manure by larvae of the black soldier fly. Journal of Food Protection, 67, 4: 685-690

Commission regulation (EU) 2017/893 of 24 May 2017 amending Annexes I and IV to regulation (EC) no 999/2001 of the European Parliament and of the council and Annexes X, XIV and XV to commission regulation (EU) no 142/2011 as regards the provisions on processed animal protein. 2017. Official Journal of the European Union, L138: 92-116

Fauna Europaea. 2016. *Hermetia illucens.* Berlin, Museum fur Naturkunde, Leibniz Institute for Evaluation and Biodiversity Science: 1 p. https://fauna-eu.org/cdm_dataportal/taxon/d9ea2161-804d-4ea2-9927-6373081010f1 (April 2018)

Filipić A. 2016. Vpliv mirovanja na mikrobioto prebavnega trakta ljudi. Magistrsko delo. Ljubljana, Biotehniška fakulteta, Študij mikrobiologije: 75 p.

Fish to 2030: prospects for fisheries and agriculture. Agriculture and environmental services discussion paper 03. 2013. Washington DC, The World Bank: 102 p. http://www.fao.org/docrep/019/i3640e/i3640e.pdf (April 2018)

Heussler C. D., Walter A., Oberkofler H., Insam H., Arthofer W., Schlick-Steiner B. C., Steiner F. M. 2018. Influence of three artificial light sources on oviposition and half-life of the black soldier fly, *Hermetia illucens* (Diptera: Stratiomyidae): improving small-scale indoor rearing. PLoS ONE, 13, 5: e0197896, doi: 10.1371/journal.pone.0197896: 10 p.

ITIS, 2017. *Hermetia illucens.* Reston, Integrated Taxonomic Information System: 2 p. https://www.itis.gov/ (April 2018)

Jeon H., Park S., Choi J., Jeong G., Lee S. B., Choi Y., Lee S. J. 2011. The intestinal bacterial community in the food waste-reducing larvae of *Hermetia illucens*. Current Microbiology, 62: 1390-1399

Jha A. K., Li J., Yuan Y., Baral N., Ai B. 2014. A review on bio-butyric acid production and its optimization. International Journal of Agriculture and Biology, 16: 1019-1024

Karak T., Bhagat R. M., Bhattacharyya P. 2012. Municipal solid waste generation, composition, and management: the world scenario. Critical Reviews in Environmental Science and Technology, 42, 15: 1509-1630

Kanaly R. A., Manzanero L. I. O., Foley G., Panneerselvam S., Macer D. 2010. Energy flow, environment and ethical implications for meat production. Bangkok, UNESCO Bangkok: 76 p. http://unesdoc.unesco.org/images/0018/001897/189774e.pdf (April 2018)

Kannan M., Suganya T., Arunprasanna V., Rameshkumar N., Krishnan M. 2015. An efficient method for extraction of genomic DNA from insect gut bacteria – culture dependent. Current Research in Microbiology and Biotechnology, 3, 1: 550-556

Kassem A., Nannipieri P. 1995. Methods in applied soil microbiology and biochemistry. 1st ed. London, Academic Press: 608 p.

Klindworth A., Pruesse E., Schweer T., Peplies J., Quast C., Horn M., Glockner F. O. 2012. Evaluation of general 16S ribosomal RNA gene PCR primers for classical and next-generation sequencing-based diversity studies. Nucleic Acid Research, 41, 1: e1, doi: 10.1093/nar/gks808: 11 p.

Kosub S. 2016. A note on the triangle inequality for the Jaccard distance. Konstanz, University of Konstanz: 5 p. https://arxiv.org/pdf/1612.02696.pdf (June 2018)

Lalander C., Diener S., Magri M. E., Zurbrugg C., Lindstrom A., Vinneras B. 2013. Faecal sludge management with larvae of the black soldier fly (*Hermetia illucens*) – from a hygiene aspect. Science of the Total Environment, 458-460: 312-318

Lalander C., Senecal J., Gros C. M., Ahrens L., Josefsson S., Wiberg K., Vinneras B. 2016. Fate of pharmaceuticals and pesticides in fly larvae composting. Science of the Total Environment, 565: 279-286

Lee Y. S., Seo S. H., Kim S. J., Yoon S. H., Hahn B. S., Sim J. S., Koo B. S., Lee C. M. 2016. Identification of a novel alkaline amylopullupanase from a gut metagenome of *Hermetia illucens*. International Journal of Biological Macromolecules, 82: 514-521

Lee C. M., Lee Y. S., Seo S. H., Yoon S. H., Kim S. J., Hahn B. S., Sim J. S., Koo B. S. 2014. Screening and characterization of a novel cellulase gene from gut microflora of *Hermetia illucens* using metagenomic library. Journal of Microbiology and Biotechnology, 24, 9: 1196-1206

Leong S. Y., Kutty S. R. M., Tan C. K., Tey L. H. 2015. Comparative study on the effect of organic waste on lauric acid produced by *Hermetia illucens* larvae via bioconversion. Journal of Engineering Science and Technology, 8: 52-63

Lever M. 1977. Carbohydrate determination with 4-hydroxybenzoic acid hydrazide (PAHBAH): effect of bismuth on the reaction. Analytical Biochemistry, 81: 21–27

Li Q., Zheng L., Cai H., Garza E., Yu Z., Zhou S. 2011a. From organic waste to biodiesel: Black soldier fly, *Hermetial illucens*, makes it feasible. Fuel, 90: 1545-1548

Li Q., Zheng L., Qiu N., Cai H., Tomberlin J.K., Yu Z. 2011b. Bioconversion of diary manure by black soldier fly (Diptera: Stratiomyidae) for biodiesel and sugar production. Waste Management, 31, 6: 1316-1320

MACHEREY-NAGEL. 2016. Genomic DNA from soil. User manual NucleoSpin Soil. 2016/Rev. 06. Duren, MACHEREY-NAGEL: 27 p. http://www.mn-net.com/Portals/8/attachments/Redakteure_Bio/Protocols/Genomic%20DNA/UM_gDNASoil.pdf (April 2018)

MACHEREY-NAGEL. 2017a. Nanocolor® Ammonium 50. Duren, MACHEREY-NAGEL GmbH & Co. KG: 1 p. ftp://ftp.mn-net.com/english/Instruction_leaflets/NANOCOLOR/985005en.PDF (April 2018)

MACHEREY-NAGEL. 2017b. Nanocolor® COD 1500. Duren, MACHEREY-NAGEL GmbH & Co. KG: 1 p. ftp://ftp.mn-net.com/english/Instruction_leaflets/NANOCOLOR/985029en.PDF (April 2018)

Mothur. 2008. Michigan, University of Michigan, Department of Microbiology and Immunology: software https://mothur.org/ (June 2018)

Newton L., Sheppard C., Watson D. W., Burtle G., Dove R. 2005. Using the black soldier fly, *Hermetia illucens*, as a value-added tool for the management of swine manure. Raleigh, North Carolina State University: 19 p. http://www.organicvaluerecovery.com/studies/studies_htm_files/bsf_value_added.pdf (June 2018)

Nguyen T. T. X., Tomberlin J. K., Vanlaerhoven S. 2015. Ability of black soldier fly (Diptera: Stratiomyidae) larvae to recycle food waste. Environmental Entomology, 44, 2: 406-410

Noble J. E., Bailey M. J. A. 2009. Quantification of protein. In: Methods in enzymology – guide to protein purification. Vol. 463. 2nd ed. Burgess R.R., Deutscher M. P. (eds.). San Diego, Academic Press: 73-96

Oliveira F., Doelle R., List R., O'Reilly J. R. 2015. Assessment of Diptera: Stratomyidae, genus *Hermetia illucens* (L., 1758) using electron microscopy. Journal of Entomology and Zoology Studies, 3, 5: 147-152

Park S. I., Chang B. S., Yoe S. M. 2014. Detection of antimicrobial substances from larvae of the black soldier fly, *Hermetia illucens* (Diptera: Stratiomyidae). Entomological Research, 44: 58-65

Parra Paz A. S., Carrejo N. S., Gomez Rodriguez C. H. 2015. Effects of larval density and feeding rates on the bioconversion of vegetable waste using black soldier fly larvae *Hermetia illucens* (L.), (Diptera: Stratomyidae). Waste and Biomass Valorization, 6, 6: 1059-1065

Pluznick J. L. 2014. Gut microbes and host physiology: what happens when you host billions of guests? Frontiers in Endocrinology, 5: 91, doi: 10.3389/fendo.2014.00091: 3 p.

Riuji Lohri C., Diener S., Zabaleta I., Mertenat A., Zurbrugg C. 2017. Treatment technologies for urban solid biowaste to create value products: a review with focus on low- and middle- income settings. Reviews in Environmental Science Biotechnology, 16: 81-130

Ruhnke I., Normant C., Campbell D. L. M., Iqbal Z., Lee C., Hinch G. N., Roberts J. 2018. Impact of on-range choice feeding with black soldier fly larvae (*Hermetia illucens*) on flock performance, egg quality, and range use of free-range laying hens. Animal Nutrition, doi: 10.1016/j.aninu.2018.03.005: 9 p. (in press)

Schloss P. D. 2008. Evaluating different approaches that test whether microbial communities have the same structure. ISME Journal, 2: 265-275

Schloss P. D., Westcott S. L., Ryabin T., Hall J. R., Hartmann M., Hollister E. B., Lesniewski R. A., Oakley B. B., Parks D. H., Robinson C. J., Sahl J. W., Stres B., Thallinger G. G., Van Horn D. J., Weber C. F. 2009. Introducing mothur: open-source, platform-independent, community-supported software for describing and comparing microbial communities. Applied and Environmental Microbiology, 75, 23: 7537-7541

Smet J. D., Wynants E., Cos P., Campenhout L. V. 2018. Microbial community dynamics during rearing of black soldier fly larvae (*Hermetia illucens*) and its impact on exploitation potential. Applied and Environmental Microbiology, 84: e02722-17, doi: 10.1128/AEM.02722-17: 15 p. (in press)

St-Hilaire S., Cranfill K., McGuire M. A., Mosley E. E., Tomberlin J. K., Newton L., Sealey W., Sheppard C., Irving S. 2007. Fish offal recycling by the black soldier fly produces a foodstaff high in omega-3 fatty acids. Journal of the World Aquaculture Society, 38, 2: 309-313

Šket R., Treichel N., Kublik S., Debevec T., Eiken O., Mekjavić I., Schloter M., Vital M., Chandler J., Tiedej J. M., Murovec B., Prevoršek Z., Likar M., Stres B. 2017. Hypoxia and inactivity related physiological changes precede or take place in absence of significant rearrangements in bacterial community structure: The PlanHab randomized trial pilot study. PLoS ONE, 12, 12: e0188556, doi: 10.1371/journal.pone0188556: 26 p.

Wagner A. O., Markt R., Puempel T., Illmer P., Insam H., Ebner C. 2017. Sample preparation, preservation, and storage for volatile fatty acids quantification in biogas plants. Engineering in Life Sciences, 17: 132-139

Wang Y. S., Shelomi M. 2017. Review of black soldier fly (*Hermetia illucens*) as animal feed and human food. Foods, 6: 91, doi: 10.3390/foods6100091: 23 p.

Wickelmaier F. 2003. An introduction to MDS. Aalborg, Aalborg University: 26 p. http://citeseerx.ist.psu.edu/viewdoc/download;jsessionid=B802976FBFCB53AD2B0AEB79CA7185DF?doi=10.1.1.495.4629&rep=rep1&type=pdf (June 2018)

Yu G., Cheng P., Chen Y., Li Y., Yang Z., Chen Y., Tomberlin J. K. 2011. Inoculating poultry manure with companion bacteria influences growth and development of black soldier fly (Diptera: Stratomyidae) larvae. Environmental Entomology, 40, 1: 30-35

Yue J. C., Clayton M. K. 2005. A similarity measure based on species proportions. Communications in Statistics-Theory and Methods, 34: 2123-2131

Zheng L., Crippen T. L., Singh B., Tarone A. M., Dowd S., Yu Z., Wood T. K., Tomberlin J. K. 2013. A survey of bacterial diversity from successive life stages of black soldier fly (Diptera: Stratomyidae) by using 16S rDNA pyrosequencing. Journal of Medical Entomology, 50, 3: 647-658

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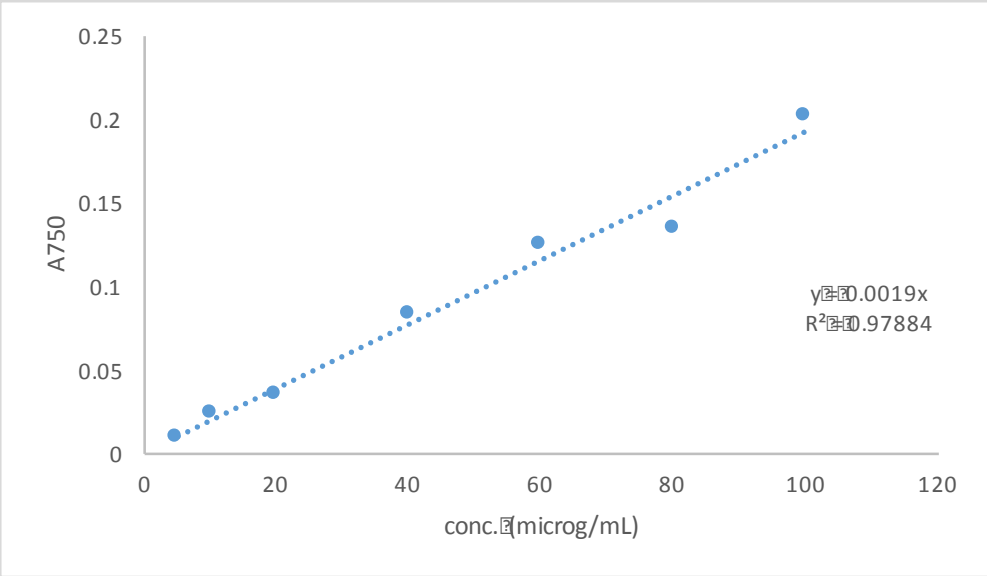
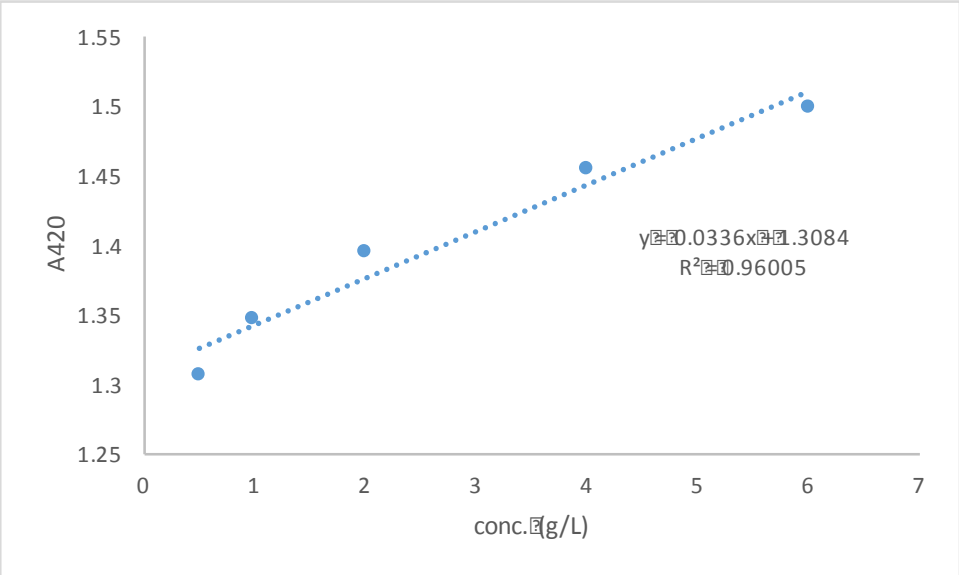
ANNEXES



Annex A: Standard curve, Lowry method.



Annex B: Standard curve for PAHBAH reducing sugar method.

Annex C: Results of substrate characterization (n=3).

Substrate		Oil residuals	Food residuals	Chicken feed	Pinien humus
Active pH		4.18±0.03	4.44±0.01	6.07±0.02	4.52±0.02
Passive pH		4.05±0.02	4.48±0.01	5.92±0.01	3.31±0.01
DS (%)		54.13±2.18	19.35±0.47	90.13±0.25	61.45±1.14
VS (%)		99.79±0.03	94.22±0.24	85.26±0.68	81.73±5.84
COD (mg/L O_2)		3657±19	8920±1394	18027±471	940±31
NH_4 (mg/L NH_4-N)		21.67±0.68	16.87±1.4	9.20±1.44	< 1.0
Total C (%)		71.03±0.29	46.65±0.44	40.96±0.69	39.83±0.57
Total N (%)		0.69628±0.10060	2.76437±0.17575	2.95390±0.08315	0.43089±0.02183
Protein (mg/mL)		0.299±0.020	2.526±0.278	3.588±0.175	1.283±0.080
Reducing sugar (g/L)		< 0.5	7.694±1.772	3.778±1.040	< 0.5
Fat (%)		29.7±6.1	5.279±0.454	3.603±0.200	0.250±0.081
Volatile fatty acids	Fo (g/mL)	0.00±0.00	0.00±0.00	4.18±0.00	0.00±0.00
	Ac (g/mL)	7.42±0.00	32.06±0.00	56.97±0.00	0.00±0.00
	Pr (g/mL)	29.73±0.00	16.06±0.00	1.11±0.00	0.78±0.00
	Bu (g/mL)	2.27±0.00	2.51±0.00	3.67±0.00	0.10±0.00
	Lac (g/mL)	0.12±0.00	18.83±0.00	35.68±0.00	0.00±0.00
	i-bu (g/mL)	2.73±0.00	0.00±0.00	0.00±0.00	0.00±0.00
	i-val (g/mL)	1.49±0.00	0.00±0.00	0.53±0.00	0.00±0.00
	Val (g/mL)	4.86±0.00	1.65±0.00	0.00±0.00	0.00±0.00

Annex D: Results of residual characterization (n=6).

Residual		Box 1-4	Box 5-8	Box 9-12
Active pH		5.74±0.13	6.46±0.06	6.26±0.05
Passive pH		6.22±0.03	4.98±0.04	6.31±0.04
DS (%)		87.11±0.80	75.43±2.38	81.00±1.58
VS (%)		80.60±0.57	97.62±0.30	88.26±0.71
COD (mg/L O_2)		5484±4272	4588±643	2728±322
NH_4 (mg/L NH_4-N)		194.88±48.85	71.83±18.84	539.00±131.83
Total C (%)		40.54±0.90	67.15±1.17	44.87±1.71
Total N (%)		3.35713±0.04951	0.70627±0.07510	4.08350±0.22657
Protein (mg/mL)		36.146±5.265	3.407±1.776	25.251±7.689
Reducing sugar (g/L)		< 0.5	< 0.5	6.851±1.416
Fat (%)		4.017±1.108	40.827±6.830	9.495±1.350
Volatile fatty acids	Fo (g/mL)	0.00±0.00	0.00±0.00	0.00±0.00
	Ac (g/mL)	10.32±1.34	1.77±0.78	42.17±8.74
	Pr (g/mL)	5.76±1.27	0.22±0.10	3.10±6.99
	Bu (g/mL)	1.34±0.19	1.19±0.26	7.88±2.98
	Lac (g/mL)	15.63±1.16	0.03±0.02	8.41±2.38
	i-bu (g/mL)	36.94±2.99	6.63±1.20	68.63±22.84
	i-val (g/mL)	8.72±0.67	3.34±0.64	31.00±8.28
	Val (g/mL)	0.00±0.00	1.25±0.15	10.72±1.98

Annex E: Growth of larvae (weight, dry and organic matter) (n=4) in the feeding experiment.

	Chicken feed – box 1-4			Oil residuals – box 5-8			Food residuals – box 9-12		
Day	Larvae weight (g)	Larvae DM (%)	Larvae OM (%)	Larvae weight (g)	Larvae DM (%)	Larvae OM (%)	Larvae weight (g)	Larvae DM (%)	Larvae OM (%)
0	0.0049 ± 0.0005	25.7326 ± 0.5866	83.4872 ± 0.7491	0.0049 ± 0.0005	25.7326 ± 0.5866	83.4872 ± 0.7491	0.0049 ± 0.0005	25.7326 ± 0.5866	83.4872 ± 0.7491
2	0.0144 ± 0.0021	25.6477 ± 1.3369	83.9910 ± 1.0802	0.0073 ± 0.0011	24.4952 ± 4.4853	89.1095 ± 1.9054	0.0159 ± 0.0024	26.7458 ± 0.8208	93.3386 ± 0.7502
4	0.0404 ± 0.0069			0.0071 ± 0.0012			0.0529 ± 0.0086		
6	0.0651 ± 0.0040	29.1253 ± 1.4178	85.9991 ± 0.5317	0.0113 ± 0.0014	31.1431 ± 1.6088	95.9566 ± 0.6222	0.0789 ± 0.0105	32.6918 ± 0.4079	95.1980 ± 0.1842
8	0.1109 ± 0.0101			0.0151 ± 0.0041			0.1430± 0.0190		
10	0.1505 ± 0.0061	34.0426 ± 1.0528	86.6412 ± 1.5944	0.0168 ± 0.0020	29.3738 ± 1.9638	99.5539 ± 0.0884	0.1819 ± 0.0272	36.2056 ± 0.3234	95.7192 ± 0.4567
12	0.1751 ± 0.0033			0.0217 ± 0.0124	33.8295 ± 6.2360	99.1989 ± 0.7771	0.1956 ± 0.0235		
14	0.179 ± 0.0119	37.423 ± 0.5629	87.415 ± 0.2322				0.2053 ± 0.0111	35.9843 ± 0.7228	95.5225 ± 0.2321
16	0.183 ± 0.0029						0.1875 ± 0.0214		
18	0.170 ± 0.0101	36.896 ± 0.2443	87.4325 ± 0.2975				0.1992 ± 0.0161	35.7121 ± 1.3374	95.5731 ± 0.3449
20							0.1641 ± 0.0168	34.8285 ± 2.5131	95.7397 ± 0.4271

Annex F: Results of pupae characterization (weight, dry and organic matter).

	Weight (g)	DM (%)	OM (%)
Box1-4	0.1447±0.0175	36.5130±0.8537	87.6525±0.6552
Box9-12	0.1691±0.0113	36.7100±0.9238	95.9569±0.5062

Annex G: Results of one-way PERMANOVA, AMOVA and HOMOVA tests for substrate sequencing data (subsampled OTU data; number of sequences: 459).

Test	One-way PERMANOVA		AMOVA				HOMOVA		
Similarity index	Bray-Curtis		jclass		thetayc		jclass	thetayc	
p-value	C-O	0.103	C-O	0.103	C-O	**0.049**	0.148	C-O	0.107
	C-F	0.101	C-F	0.114	C-F	0.067		C-F	0.102
	O-F	0.099	O-F	0.114	O-F	0.095		O-F	0.396

Annex H: OTU data on genera level, presented in larvae gut samples.

	Acetobacter	*Achromobacter*	*Acidaminococcaceae*_unclassified	*Acinetobacter*	*Actinobacteria*_unclassified	*Actinomycetales*_unclassified	*Anaerolineaceae*_unclassified	*Anaerosinus*	*Arcobacter*	*Bacillales*_unclassified	*Bacilli*_unclassified	*Bacteria*_unclassified
C0.1	0	2	0	145	0	860	18	1	0	3	9	32
C0.2	0	5	1	204	0	960	14	0	0	1	11	20
C0.3	0	5	0	183	0	1057	11	1	0	1	15	8
C2.1	0	1	1	4160	0	161	7	0	0	1	0	8
C2.2	0	0	0	128	0	328	11	0	0	2	1	14
C2.3	0	1	0	834	1	667	12	0	0	1	1	16
C10.1	0	0	0	2	0	3052	11	0	0	0	32	14
C10.2	0	2	0	13	0	5967	11	0	2	0	25	15
C10.3	0	2	0	13	0	5342	12	0	0	0	17	21
C14.1	0	0	0	6	1	2410	18	0	0	0	0	24
C14.2	0	1	0	4	0	4263	12	0	0	0	2	21
C14.3	0	0	0	5	0	1585	14	2	0	0	3	13
C18.1	0	0	0	2	0	2245	16	0	0	0	27	22
C18.2	0	0	0	0	0	1464	22	0	1	0	14	25
C18.3	0	1	0	1	1	1539	18	1	0	1	5	24
F2.1	0	0	1	10	0	1188	13	1	1	1	2	11
F2.2	0	0	0	7	0	441	13	3	0	1	0	21
F2.3	0	0	0	11	0	101	8	0	0	1	0	9
F10.1	0	0	0	1	0	6413	10	0	0	0	2	15
F10.2	0	0	0	0	0	661	16	1	0	0	1	20
F10.3	0	0	0	5	1	1703	6	2	0	0	0	13
F14.1	0	0	0	2	1	810	9	1	0	0	1	12
F14.2	0	0	0	6	0	274	11	0	0	1	0	11
F14.3	0	0	0	2	1	2575	40	0	0	0	2	31
F20.1	0	0	0	0	0	559	10	0	0	0	1	14
F20.2	0	0	0	1	1	430	6	0	0	0	0	11
F20.3	0	0	0	14	1	268	13	1	0	1	0	16
O2.1	0	0	0	6493	0	2001	21	0	1	32	3	30
O2.2	0	0	0	6482	0	1572	17	3	1	13	0	23
O2.3	0	0	1	7710	1	332	12	0	0	7	0	15
O10.1	9	0	1	230	0	57	15	2	0	0	0	17
O10.2	16	0	0	332	0	97	16	1	0	1	0	14
O10.3	59	0	0	329	0	181	9	0	0	0	0	15
O12.1	14	0	2	112	1	66	14	0	1	0	0	30
O12.2	4	0	0	35	2	78	14	0	1	0	0	22
O12.3	8	0	4	88	0	77	12	0	0	2	0	17
TOTAL	**110**	**20**	**11**	**27570**	**12**	**51784**	**492**	**20**	**8**	**70**	**174**	**644**

continued

continuation of Annex H

	*Bacteroidales*_unclassified	*Bacteroidetes*_unclassified	*Bellilinea*	*Bifidobacterium*	*Bosea*	*Brachybacterium*	*Bradyrhizobium*	*Brevibacterium*	*Brucellaceae*_unclassified	*Burkholderia*	*Camelimonas*	*Candidatus*_*Cloacamonas*_unclassified
C0.1	3	24	0	0	8	0	0	0	64	0	2	17
C0.2	6	19	1	0	11	0	2	0	73	0	7	7
C0.3	1	12	0	0	6	0	0	1	62	0	6	3
C2.1	0	16	1	0	0	0	0	3	6	0	0	1
C2.2	0	21	1	0	0	0	0	0	0	1	0	8
C2.3	0	30	0	0	0	0	0	2	9	0	0	13
C10.1	0	17	0	0	0	0	0	3	8	0	0	9
C10.2	0	19	0	0	0	0	0	7	17	0	0	15
C10.3	0	13	0	0	0	0	0	4	19	0	0	5
C14.1	0	21	1	0	0	1	0	0	0	0	0	11
C14.2	0	19	1	1	0	9	0	2	2	1	0	11
C14.3	0	7	0	1	0	1	0	5	1	0	0	2
C18.1	0	21	1	0	0	0	0	0	0	0	0	10
C18.2	0	17	0	0	0	0	1	2	2	1	0	15
C18.3	0	22	1	0	0	1	0	1	2	0	0	7
F2.1	10	13	1	0	0	0	1	0	0	0	0	4
F2.2	0	34	0	0	0	0	0	0	1	2	0	3
F2.3	0	10	1	0	0	0	0	0	1	0	0	5
F10.1	1	16	0	0	0	0	0	0	0	0	0	8
F10.2	5	17	1	0	0	0	0	0	0	0	0	17
F10.3	0	12	0	0	0	0	0	0	0	0	0	3
F14.1	6	11	0	0	0	0	0	0	0	0	0	6
F14.2	0	26	0	0	0	0	0	4	0	0	0	7
F14.3	0	51	1	0	0	0	8	1	2	5	0	26
F20.1	0	10	0	0	0	12	0	94	0	0	0	4
F20.2	5	15	0	0	0	0	0	2	0	0	0	2
F20.3	0	23	1	0	0	0	0	0	0	0	0	7
O2.1	0	23	2	5	0	0	0	0	1	1	0	22
O2.2	2	15	2	5	0	0	0	0	0	0	0	6
O2.3	2	11	0	1	0	0	0	0	0	0	0	9
O10.1	2	22	0	2	0	0	1	1	0	0	0	4
O10.2	0	14	0	8	0	0	1	0	0	1	0	2
O10.3	8	11	1	2	0	0	0	1	0	0	0	4
O12.1	0	25	0	13	0	0	0	1	0	1	0	10
O12.2	2	19	0	1	0	0	0	0	0	0	0	10
O12.3	0	23	0	4	0	0	0	2	0	2	0	14
TOTAL	**53**	**679**	**17**	**43**	**25**	**24**	**14**	**136**	**270**	**15**	**15**	**307**

continued

continuation of Annex H

	*Candidatus_ Saccharibacteria_* unclassified	*Chryseobacterium*	*Clostridiaceae_1_* unclassified	*Clostridiales_Incertae_ Sedis_XI_* unclassified	*Clostridiales_* unclassified	*Clostridium_sensu_ stricto*	*Clostridium_XIVa*	*Comamonadaceae_* unclassified	*Comamonas*	*Corynebacterium*	*Desulfovibrio*	*Devosia*
C0.1	2	1050	1	1	1	8	4	2	5	2	0	1
C0.2	4	868	0	2	0	3	3	11	6	1	0	11
C0.3	8	688	4	0	6	3	3	7	9	4	0	4
C2.1	0	1278	0	0	0	2	0	1	1	12	0	0
C2.2	0	2	0	2	0	0	0	0	0	0	0	0
C2.3	0	666	0	0	0	5	0	0	0	19	0	0
C10.1	0	7	0	1	0	2	0	0	0	1	0	0
C10.2	0	26	0	1	5	5	0	0	0	1	0	0
C10.3	0	38	0	0	1	1	2	0	0	2	0	0
C14.1	0	1	0	0	1	2	1	1	0	3	0	0
C14.2	0	4	0	1	9	0	0	0	0	19	1	0
C14.3	0	1	0	1	0	0	0	0	1	40	0	0
C18.1	0	0	0	0	0	2	0	0	0	4	0	0
C18.2	0	2	0	2	0	1	0	0	0	2	0	0
C18.3	0	0	0	0	0	0	0	0	0	19	0	0
F2.1	0	1	0	0	1	2	0	0	1	1	1	0
F2.2	0	0	0	3	3	1	0	0	0	1	0	0
F2.3	0	0	0	0	4	0	0	1	0	0	0	0
F10.1	0	0	0	2	2	2	0	0	0	1	0	0
F10.2	0	0	0	1	0	4	0	0	0	0	0	0
F10.3	0	0	0	0	0	2	0	0	0	3	0	0
F14.1	0	0	0	1	1	2	1	1	0	11	0	0
F14.2	0	1	0	1	1	1	0	0	1	52	0	0
F14.3	0	0	0	3	0	12	0	28	3	4	0	0
F20.1	0	1	0	0	3	1	0	1	0	86	0	0
F20.2	0	1	0	0	0	0	0	0	0	130	0	0
F20.3	0	3	0	1	1	5	0	0	0	4	0	0
O2.1	0	1	1	0	7	6	106	2	0	0	0	0
O2.2	0	4	0	0	1	6	2	1	1	1	0	0
O2.3	0	1	0	2	2	6	83	1	1	2	0	0
O10.1	0	0	0	1	9	38	0	0	6	2	1	0
O10.2	0	2	0	1	1	57	129	0	0	2	0	0
O10.3	0	0	0	0	1	114	1	1	9	1	1	0
O12.1	0	1	0	2	4	23	55	12	11	2	2	0
O12.2	0	0	0	0	1	13	0	0	0	1	2	0
O12.3	0	0	1	1	7	35	39	17	13	2	3	0
TOTAL	**14**	**4647**	**7**	**30**	**72**	**364**	**429**	**87**	**68**	**435**	**11**	**16**

continued

continuation of Annex H

	Dialister	*Dysgonomonas*	*Empedobacter*	*Enterobacteriaceae*_ unclassified	*Enterococcus*	*Firmicutes*_unclassified	*Gammaproteobacteria*_ unclassified	*Intestinibacter*	*Kaistia*	*Lachnospiraceae*_ unclassified	*Lactobacillales*_ unclassified	*Lactobacillus*
C0.1	0	4955	347	12827	11895	0	21	2	3	5	20	5
C0.2	0	5168	180	13479	10462	0	16	0	2	56	39	0
C0.3	1	9329	174	12787	9151	0	16	1	5	90	34	4
C2.1	1	1319	2921	16506	3501	0	150	1	0	0	11	7
C2.2	0	15899	20	13886	3215	0	742	0	0	0	1	15
C2.3	0	6661	1331	17309	5227	0	540	0	0	0	10	5
C10.1	0	12782	1	2859	13523	2	2814	0	0	0	25	0
C10.2	0	9829	7	2991	15394	1	994	0	0	2	71	0
C10.3	0	4824	3	1206	20900	2	2300	1	0	0	46	2
C14.1	0	18284	0	5855	6062	6	1568	0	0	1	5	0
C14.2	3	2834	0	4521	23567	0	523	1	0	0	6	14
C14.3	1	10154	2	2126	20688	0	806	1	0	169	20	2
C18.1	0	13420	0	3921	15537	3	719	0	0	0	18	1
C18.2	0	12674	1	11143	9095	0	892	0	0	0	10	1
C18.3	0	1633	0	13154	19492	1	36	0	0	0	44	0
F2.1	0	5296	1	17792	4697	0	65	0	0	0	14	231
F2.2	0	28955	0	1039	1047	0	17	0	0	0	0	2100
F2.3	0	3776	0	688	331	0	2	0	0	2	3	262
F10.1	0	2239	0	1133	15927	0	4011	0	0	0	56	8
F10.2	0	4735	1	1606	3068	1	428	1	0	0	29	306
F10.3	0	9456	0	5574	14948	0	2811	0	0	0	21	96
F14.1	0	1802	1	1530	3897	0	582	1	0	0	26	92
F14.2	0	190	0	130	5493	0	548	0	0	0	36	14
F14.3	1	603	0	4704	8462	2	1747	2	0	0	1	63
F20.1	0	759	0	198	3013	0	208	0	0	0	10	0
F20.2	1	136	3	455	2345	0	511	0	0	0	5	1
F20.3	0	199	0	73	1192	0	367	1	0	0	3	0
O2.1	0	10096	0	7928	3490	2	82	0	0	0	7	0
O2.2	0	8365	4	10199	5570	0	655	1	0	0	6	4
O2.3	1	15660	1	7839	835	1	32	0	0	0	0	1
O10.1	5	3369	1	967	184	0	16	2	0	1	1	36
O10.2	0	12029	2	719	257	0	9	0	0	17	1	91
O10.3	1	1567	0	518	844	0	232	1	0	0	3	31
O12.1	1	4658	7	1084	466	0	8	1	0	0	1	37
O12.2	3	3956	0	5728	362	0	13	0	0	0	1	21
O12.3	2	5763	3	1144	487	0	5	0	0	0	3	29
TOTAL	**21**	**253374**	**5011**	**205618**	**264624**	**21**	**24486**	**17**	**10**	**343**	**587**	**3479**

continued

continuation of Annex H

	Lactococcus	*Leuconostoc*	*Megasphaera*	*Methylocystaceae*_ unclassified	*Methylogaea*	*Methylophilus*	*Microbacteriaceae*_ unclassified	*Microbacterium*	*Micrococcaceae*_ unclassified	*Morganella*	*Ochrobactrum*	*Orbaceae*_unclassified
C0.1	8	4	0	0	0	0	0	16	4	53	37	4
C0.2	0	3	2	0	0	0	0	14	1	97	60	3
C0.3	20	0	1	0	0	0	1	17	2	113	29	1
C2.1	18	0	0	0	0	0	0	1	0	32	2	26
C2.2	67	0	0	0	0	1	0	1	1	25	0	804
C2.3	0	1	0	0	0	1	0	4	7	44	0	658
C10.1	0	0	0	0	0	0	0	4	2	289	3	955
C10.2	0	2	0	0	0	0	3	5	6	92	6	398
C10.3	0	3	0	0	0	0	0	0	6	76	4	1665
C14.1	1	14	0	0	0	1	1	0	4	394	0	1138
C14.2	67	0	2	0	0	1	5	3	41	185	0	443
C14.3	13	0	2	0	0	0	6	3	22	121	0	634
C18.1	23	0	0	0	0	0	0	3	0	83	0	595
C18.2	0	0	0	0	0	0	1	0	2	157	0	1116
C18.3	0	0	0	0	0	0	0	0	2	665	1	79
F2.1	0	13	0	3	12	0	0	0	0	212	0	39
F2.2	3	31	1	0	0	1	2	1	0	816	0	8
F2.3	16	1	1	1	0	0	1	0	0	42	1	76
F10.1	14	0	0	0	0	0	0	0	1	461	0	4779
F10.2	0	0	0	0	0	0	0	1	0	23515	0	141
F10.3	15	1	0	0	0	0	0	0	0	462	0	880
F14.1	0	4	0	1	0	0	0	1	1	22286	0	3172
F14.2	1	1	0	0	0	0	0	0	0	27856	0	502
F14.3	0	1	2	0	0	2	0	0	1	1755	0	13445
F20.1	16	0	1	0	0	0	0	0	3	31146	0	134
F20.2	9	0	0	0	0	0	0	0	0	31758	0	382
F20.3	0	2	0	0	0	0	0	0	0	34287	0	282
O2.1	0	2	4	0	3	0	1	5	2	553	0	15
O2.2	1	13	5	0	0	2	0	2	1	1400	0	39
O2.3	0	9	2	1	0	0	0	0	3	983	0	1
O10.1	35	1	15	0	4	0	1	0	3	25438	1	4
O10.2	14	0	12	0	0	0	0	2	0	21805	0	4
O10.3	8	0	1	0	0	0	0	0	0	24968	0	124
O12.1	0	0	14	0	0	0	0	1	1	26962	0	100
O12.2	0	0	13	0	0	0	0	0	0	22762	0	187
O12.3	0	1	19	0	0	0	0	0	0	25378	0	56
TOTAL	**349**	**107**	**97**	**6**	**19**	**9**	**22**	**84**	**116**	**327271**	**144**	**32889**

continued

continuation of Annex H

	Paenalcaligenes	*Paenibacillus*	*Paenochrobactrum*	*Pectinatus*	*Pediococcus*	*Planctomycetaceae_* unclassified	*Porphyromonadaceae_*un classified	*Prevotella*	*Propionibacterium*	*Proteus*	*Providencia*	*Pseudomonas*
C0.1	4	2	0	0	0	0	1699	0	0	0	944	3
C0.2	3	2	3	0	0	0	2124	3	2	0	1122	2
C0.3	2	3	2	0	0	0	1073	1	0	0	794	0
C2.1	0	0	0	0	0	1	242	3	0	0	149	1
C2.2	0	0	0	0	0	1	265	0	1	0	352	2
C2.3	1	0	0	0	0	0	760	0	2	1	598	1
C10.1	0	0	0	0	0	0	326	0	0	14	41	0
C10.2	0	0	0	0	0	0	667	0	0	2	105	0
C10.3	0	0	1	0	0	0	167	0	0	1	14	0
C14.1	0	0	0	0	0	0	871	2	0	0	124	0
C14.2	0	0	0	0	0	0	19	25	1	0	80	0
C14.3	0	0	0	0	0	0	117	5	0	0	30	0
C18.1	0	0	0	0	0	0	74	1	0	8	69	0
C18.2	0	0	0	0	0	0	26	0	0	61	70	0
C18.3	0	0	1	0	0	2	2	0	0	0	70	0
F2.1	7	1	0	0	9	0	4190	5	1	0	1816	2
F2.2	0	0	0	0	19	1	200	0	0	0	1052	2
F2.3	3	0	0	0	7	0	153	4	0	0	30699	3
F10.1	0	0	0	0	0	0	1064	0	0	0	677	0
F10.2	0	0	0	0	1	0	584	0	0	0	147	0
F10.3	0	0	0	0	1	1	257	2	0	0	353	0
F14.1	0	0	0	0	0	0	1060	0	0	2	774	2
F14.2	0	0	0	0	0	0	1492	0	0	0	97	0
F14.3	0	0	0	0	1	0	1874	2	4	0	589	10
F20.1	0	0	0	0	0	1	301	2	0	0	191	0
F20.2	0	0	0	0	0	0	455	2	0	0	186	0
F20.3	0	0	0	0	0	0	5	3	0	0	59	0
O2.1	11	1	0	0	0	0	4001	2	0	0	1626	1
O2.2	58	0	0	0	0	0	1793	0	0	0	455	1
O2.3	0	0	0	1	0	0	2342	15	2	0	818	1
O10.1	0	0	0	1	0	1	5467	20	0	0	456	4
O10.2	0	0	0	0	0	0	498	1	0	0	569	0
O10.3	0	0	0	1	0	0	1270	3	1	0	6487	2
O12.1	1	0	0	3	0	1	762	15	0	0	1806	4
O12.2	0	0	0	1	0	0	815	7	0	0	2452	1
O12.3	1	0	0	1	0	0	1063	8	0	0	1736	7
TOTAL	**91**	**9**	**7**	**8**	**38**	**9**	**38078**	**131**	**14**	**89**	**57607**	**49**

continued

continuation of Annex H

	*Rhizobiales*_unclassified	*Rhodobacteraceae*_unclassified	*Romboutsia*	*Sedimentibacter*	*Selenomonas*	*Shinella*	*Smithella*	*Solirubrobacterales*_unclassified	*Sphingobacterium*	*Sphingomonas*	*Staphylococcus*	*Stenotrophomonas*
C0.1	56	2	6	0	0	29	7	0	112	1	0	37
C0.2	85	3	3	2	1	43	0	0	151	0	0	35
C0.3	89	6	2	1	3	35	1	0	113	2	0	51
C2.1	44	0	2	0	3	1	1	0	478	0	0	57
C2.2	0	0	2	1	1	0	1	1	2	2	0	1
C2.3	30	1	7	2	0	0	4	2	45	0	14	18
C10.1	24	5	7	0	0	1	4	0	1	0	0	3
C10.2	19	2	3	1	0	2	2	0	27	0	0	32
C10.3	21	5	2	2	0	4	5	0	33	0	0	19
C14.1	0	0	5	5	1	0	3	1	1	0	0	1
C14.2	2	0	3	2	6	0	2	0	8	0	8	4
C14.3	0	2	2	2	2	0	2	0	0	0	24	4
C18.1	2	0	2	0	0	0	0	1	0	0	1	0
C18.2	5	0	1	3	0	0	3	0	2	0	0	0
C18.3	2	0	4	3	0	0	4	0	1	0	0	2
F2.1	1	0	3	3	4	0	2	0	1	0	0	0
F2.2	2	1	6	1	1	0	4	1	1	0	3	0
F2.3	3	0	0	1	2	0	2	0	0	0	0	0
F10.1	0	0	4	0	0	0	1	0	0	0	0	0
F10.2	0	0	5	0	0	0	3	0	0	0	0	0
F10.3	0	0	3	0	2	0	0	0	0	0	0	0
F14.1	0	0	4	2	0	0	3	0	0	0	2	0
F14.2	0	0	4	2	0	0	1	0	0	0	13	0
F14.3	2	0	12	3	4	0	11	0	1	7	162	0
F20.1	0	0	4	1	0	1	0	0	1	0	56	0
F20.2	0	0	2	2	2	0	2	0	0	0	1	0
F20.3	0	0	3	6	3	0	1	0	0	0	11	0
O2.1	2	0	6	1	0	0	5	0	1	0	0	0
O2.2	2	0	7	6	2	0	1	0	0	1	0	0
O2.3	0	0	3	3	21	0	3	0	0	0	0	0
O10.1	1	0	4	0	47	0	1	1	1	0	1	7
O10.2	0	0	2	3	14	0	0	0	3	1	1	5
O10.3	0	0	3	0	3	0	2	0	3	0	0	2
O12.1	0	0	0	1	45	0	3	0	7	0	5	6
O12.2	0	0	6	0	34	1	3	0	0	0	1	0
O12.3	0	0	10	3	35	0	5	0	6	1	10	16
TOTAL	**392**	**27**	**142**	**62**	**236**	**117**	**92**	**7**	**999**	**15**	**313**	**300**

continued

continuation of Annex H

	Streptococcus	*Synergistaceae*_ unclassified	*Syntrophomonas*	*Tepidiphilus*	*Terrisporobacter*	*Tolumonas*	*Turicibacter*	*Vagococcus*	*Veillonella*	*Veillonellaceae*_ unclassified	*Verrucomicrobia*_u nclassified	*Weissella*	*Xanthomonas*
C0.1	1	3	2	0	3	0	0	200	0	0	3	1226	0
C0.2	1	0	3	0	1	0	1	334	0	1	3	1049	0
C0.3	0	0	0	0	0	0	0	222	0	2	1	551	0
C2.1	0	2	0	0	0	0	0	12	0	4	0	5691	0
C2.2	0	0	1	0	0	0	0	12	1	2	3	1000	1
C2.3	0	0	0	0	0	0	1	37	0	0	1	1259	0
C10.1	0	1	0	0	0	0	0	0	0	0	1	10	0
C10.2	0	2	1	0	0	0	1	2	0	0	1	55	0
C10.3	0	0	1	0	1	0	0	0	0	0	5	32	0
C14.1	1	2	1	0	0	0	0	2	0	0	1	4	0
C14.2	0	1	0	0	0	0	0	2	2	8	1	71	3
C14.3	0	2	0	0	0	0	0	4	1	2	0	205	1
C18.1	0	1	2	0	0	0	0	0	0	0	1	13	0
C18.2	0	0	0	0	2	0	0	1	0	0	2	6	0
C18.3	0	1	1	0	1	0	0	1	0	0	2	11	0
F2.1	0	1	0	0	1	0	0	644	1	7	2	504	0
F2.2	5	1	1	0	1	0	0	24	0	1	1	958	0
F2.3	0	0	0	0	0	0	0	4	0	0	0	623	0
F10.1	0	1	0	0	0	0	0	1	0	1	0	3	0
F10.2	0	3	1	0	1	0	0	1	0	0	3	1536	0
F10.3	0	0	0	0	0	0	0	11	0	3	1	208	0
F14.1	0	2	0	0	0	0	0	2	0	0	0	733	0
F14.2	0	1	1	0	0	0	0	0	0	0	1	85	0
F14.3	0	4	0	6	0	0	0	27	0	0	8	450	0
F20.1	0	0	0	0	0	0	0	10	0	0	1	11	0
F20.2	0	0	2	0	0	0	0	2	0	1	0	9	0
F20.3	0	1	1	0	0	0	0	1	0	0	5	1	0
O2.1	1	2	1	0	1	0	1	168	0	7	4	36	0
O2.2	0	2	1	0	1	0	1	54	0	4	2	27	0
O2.3	0	0	2	0	0	1	0	27	0	12	1	18	0
O10.1	0	0	2	1	1	2	2	281	3	31	0	10	0
O10.2	0	3	0	1	0	1	0	69	1	7	0	15	1
O10.3	1	1	0	0	1	0	0	7	1	11	1	13	0
O12.1	0	2	0	0	0	0	0	399	2	37	2	20	0
O12.2	0	0	0	0	0	2	1	230	1	22	2	20	0
O12.3	1	0	0	0	0	0	1	620	4	30	2	23	0
TOTAL	**11**	**39**	**24**	**8**	**15**	**6**	**9**	**3411**	**17**	**193**	**61**	**16486**	**6**

Annex I: OTU data on general level, presented in substrate samples.

	C1	C2	C3	F1	F2	O1	O2	O3	TOTAL
Acetobacter	0	1	0	0	0	0	0	1	**2**
Achromobacter	0	0	1	0	0	0	0	0	**1**
*Acidaminococcaceae_* unclassified	0	0	0	0	0	0	5	2	**7**
Acinetobacter	16	3	4	1	3	1	0	0	**28**
*Actinobacteria_*unclassified	1	0	1	0	0	0	0	0	**2**
*Actinomycetales_*unclassified	12	3	10	0	0	0	0	1	**26**
*Actinopolysporaceae_* unclassified	0	1	0	0	0	0	0	0	**1**
Aeribacillus	0	0	1	0	0	0	0	0	**1**
*Alcaligenaceae_*unclassified	0	3	0	0	0	0	0	0	**3**
Alcaligenes	0	1	0	0	0	0	0	0	**1**
Alistipes	0	1	0	0	0	0	0	0	**1**
Alloprevotella	0	1	0	0	0	0	0	0	**1**
Anaerolinea	1	0	1	0	0	0	0	0	**2**
*Anaerolineaceae_*unclassified	20	19	34	1	0	0	0	0	**74**
Anaeromyxobacter	0	0	1	0	0	0	0	0	**1**
Anaerosinus	0	2	2	0	1	0	1	0	**6**
Aquabacterium	0	0	2	0	0	0	0	0	**2**
*Armatimonadetes_gp2_* unclassified	0	1	1	0	0	0	0	0	**2**
Arsenicicoccus	0	1	1	0	0	0	0	0	**2**
Arthrobacter	0	0	1	0	0	0	0	0	**1**
Aureimonas	0	0	1	0	0	0	0	0	**1**
Azonexus	0	0	0	0	0	1	2	0	**3**
*Bacillaceae_1_*unclassified	0	1	1	0	0	0	0	0	**2**
*Bacillaceae_2_*unclassified	2	0	0	0	0	0	0	0	**2**
*Bacillales_*unclassified	3	3	7	0	0	0	0	0	**13**
*Bacteria_*unclassified	43	49	73	4	0	5	5	6	**185**
*Bacteroidetes_*unclassified	40	38	54	0	5	2	6	6	**151**
*Betaproteobacteria_* unclassified	0	0	4	1	0	0	0	0	**5**
Bifidobacterium	0	0	0	0	0	1	3	2	**6**
Blastococcus	2	0	0	0	0	0	0	0	**2**
*Bradyrhizobiaceae_* unclassified	0	0	1	0	0	0	0	0	**1**
*Brevibacteriaceae_* unclassified	1	0	0	0	0	0	0	0	**1**
Brevibacterium	3	2	3	0	0	0	0	0	**8**
Brevundimonas	0	0	1	0	0	0	0	0	**1**
*Brucellaceae_*unclassified	2	0	0	0	0	0	0	0	**2**
Burkholderia	2	0	0	0	0	0	0	0	**2**
Camelimonas	0	0	1	0	0	0	0	0	**1**
*Candidatus_Cloacamonas_* unclassified	5	24	22	0	0	1	0	0	**52**
*Candidatus_Hydrogenedens_* unclassified	2	3	2	0	0	0	0	0	**7**
Cellulomonas	0	1	0	0	0	0	0	0	**1**
Chitinophaga	0	1	0	0	0	0	0	0	**1**
*Chloroflexi_*unclassified	2	2	3	0	0	0	0	0	**7**
Chryseobacterium	5	4	1	0	0	0	0	0	**10**
Cloacibacterium	1	0	0	0	0	0	1	0	**2**

continued

continuation of Annex I

	C1	C2	C3	F1	F2	O1	O2	O3	TOTAL
*Clostridia*_unclassified	1	0	1	0	0	0	0	0	**2**
*Clostridiales_Incertae_Sedis_XI_* unclassified	2	3	4	0	0	0	0	0	**9**
*Clostridiales*_unclassified	1	1	2	0	0	1	4	0	**9**
Clostridium_sensu_stricto	7	8	9	2	0	14	10	11	**61**
Clostridium_XlVa	2	0	1	0	0	0	0	0	**3**
*Comamonadaceae*_unclassified	5	4	4	0	0	0	0	0	**13**
Comamonas	0	1	2	0	0	1	0	1	**5**
Corynebacterium	1	1	1	0	0	0	0	0	**3**
Defluviitoga	1	0	0	0	0	0	0	0	**1**
Desulfovibrio	0	0	0	0	0	2	0	1	**3**
*Desulfuromonadaceae*_unclassified	1	0	0	0	0	0	0	0	**1**
*Desulfuromonadales*_unclassified	0	1	0	0	0	0	0	0	**1**
Devosia	0	0	2	0	0	0	0	0	**2**
Dialister	2	3	0	0	0	23	29	36	**93**
Dietzia	0	0	1	0	0	0	0	0	**1**
Dyadobacter	1	0	0	0	0	0	0	0	**1**
Dysgonomonas	42	35	30	0	0	0	1	0	**108**
Elusimicrobium	0	0	0	0	0	0	0	1	**1**
Empedobacter	0	2	0	0	1	0	0	0	**3**
*Enterobacteriaceae*_unclassified	227	180	229	12	6	11	6	8	**679**
Enterococcus	24	24	36	0	0	2	0	0	**86**
Epilithonimonas	1	0	1	0	0	0	0	0	**2**
Fervidobacterium	0	0	1	0	0	0	0	0	**1**
*Firmicutes*_unclassified	1	4	3	0	0	0	0	0	**8**
*Flavobacteriaceae*_unclassified	0	0	1	0	0	0	0	0	**1**
Flavobacterium	0	1	0	0	0	0	0	0	**1**
Frigoribacterium	0	1	0	0	0	0	0	0	**1**
Frondihabitans	2	2	1	0	0	0	0	0	**5**
Gaiella	1	0	0	0	0	0	0	0	**1**
*Gammaproteobacteria*_unclassified	2	1	1	0	0	9	5	14	**32**
Gemmobacter	1	0	1	0	0	0	0	0	**2**
Geobacillus	0	0	3	0	0	0	0	0	**3**
*Gilvimarinus*_unclassified	0	1	0	0	0	0	0	0	**1**
Globicatella	18	1	0	0	0	0	0	0	**19**
Glycomyces	0	0	1	0	0	0	0	0	**1**
*Gp10*_unclassified	0	0	1	0	0	0	0	0	**1**
*Gp18*_unclassified	0	0	1	0	0	0	0	0	**1**
*Gp6*_unclassified	1	0	0	0	0	0	0	0	**1**
Gulbenkiania	0	0	0	0	0	0	1	3	**4**
Hydrogenibacillus	0	2	0	0	0	0	0	0	**2**
Hydrogenobacter	0	0	1	0	0	0	0	0	**1**
Hymenobacter	1	4	0	0	0	0	0	0	**5**
Intestinibacter	0	1	3	0	0	0	0	0	**4**
Janthinobacterium	9	4	7	2	2	0	0	0	**24**

continued

continuation of Annex I

	C1	C2	C3	F1	F2	O1	O2	O3	TOTAL
Jiangella	1	0	0	0	0	0	0	0	**1**
Kineococcus	0	1	2	0	0	0	0	0	**3**
Kocuria	1	0	0	0	0	1	0	0	**2**
*Lachnospiraceae*_unclassified	1	1	0	0	0	1	0	0	**3**
*Lactobacillales*_unclassified	0	6	0	0	1	0	0	0	**7**
Lactobacillus	15	6	0	36	35	2	1	7	**102**
Lactococcus	51	3	3	41	42	1	2	1	**144**
Leadbetterella	0	0	2	0	0	0	0	0	**2**
Leuconostoc	10	17	13	630	678	0	0	0	**1348**
Litorilinea	0	1	0	0	0	0	0	0	**1**
Luteibacter	1	0	0	0	0	0	0	0	**1**
Luteimonas	1	0	0	0	0	0	0	0	**1**
Luteococcus	0	1	3	0	0	0	0	0	**4**
Megasphaera	0	8	1	0	0	53	56	63	**181**
Mesotoga	0	1	2	0	0	0	0	0	**3**
Methylobacterium	3	6	10	0	0	0	0	0	**19**
*Microbacteriaceae*_unclassified	12	12	11	0	0	0	0	0	**35**
Microbacterium	2	7	5	0	0	0	0	0	**14**
*Micrococcaceae*_unclassified	0	1	0	0	0	0	0	0	**1**
*Micromonosporaceae*_unclassified	0	2	0	0	0	0	0	0	**2**
Microvirga	0	2	0	0	0	0	0	0	**2**
Morganella	39	67	46	4	2	1	2	1	**162**
Myroides	1	0	2	0	0	0	0	0	**3**
*Neisseriaceae*_unclassified	0	0	0	0	0	0	0	1	**1**
Neorhizobium	1	4	3	0	0	0	0	0	**8**
Nocardioides	1	0	1	0	0	0	0	0	**2**
Nocardiopsis	0	2	1	0	0	0	0	0	**3**
Oceanobacillus	0	1	0	0	0	0	0	0	**1**
*Orbaceae*_unclassified	7	5	13	0	0	0	0	0	**25**
Paenalcaligenes	0	1	0	0	0	0	0	0	**1**
Paenibacillus	11	8	9	0	0	0	0	0	**28**
Pectinatus	0	0	0	0	0	1	5	2	**8**
Pediococcus	2	0	0	0	0	0	0	0	**2**
Pedobacter	4	3	0	1	0	0	0	0	**8**
Photobacterium	0	0	1	7	7	0	0	0	**15**
*Planctomycetaceae*_unclassified	3	2	1	0	0	0	0	0	**6**
*Planococcaceae*_unclassified	1	4	0	0	0	0	0	0	**5**
*Porphyromonadaceae*_unclassified	7	3	0	1	0	0	0	1	**12**
Prevotella	1	36	5	0	0	305	328	306	**981**
*Propionibacteriaceae*_unclassified	1	3	0	0	0	0	0	0	**4**
Propionibacterium	0	2	1	0	0	0	0	0	**3**
Propionispira	0	0	0	0	0	1	1	1	**3**
Propionispora	0	0	0	0	0	0	0	1	**1**
*Proteobacteria*_unclassified	2	0	0	0	0	0	0	0	**2**

continued

continuation of Annex I

	C1	C2	C3	F1	F2	O1	O2	O3	TOTAL
Providencia	14	6	5	2	0	0	0	0	**27**
Pseudokineococcus	0	0	1	0	0	0	0	0	**1**
Pseudomonas	29	22	24	36	26	0	0	0	**137**
Pseudonocardia	2	1	2	0	0	0	0	0	**5**
*Pseudonocardiaceae*_unclassified	1	11	2	0	0	0	0	0	**14**
*Rhizobiales*_unclassified	1	0	0	0	0	0	0	0	**1**
Rhizobium	1	0	0	0	0	0	0	0	**1**
*Rhodobacteraceae*_unclassified	1	0	1	0	0	0	0	0	**2**
Rhodococcus	0	4	2	0	0	0	0	0	**6**
*Rhodocyclaceae*_unclassified	1	0	1	0	0	0	0	0	**2**
*Rhodospirillales*_unclassified	0	1	0	0	0	0	0	0	**1**
Romboutsia	6	12	8	2	0	0	0	0	**28**
Roseomonas	1	0	0	0	0	0	0	0	**1**
*Ruminococcaceae*_unclassified	0	3	1	0	0	3	4	11	**22**
Saccharopolyspora	2	0	3	0	0	0	0	0	**5**
Sanguibacter	4	4	0	0	0	0	0	0	**8**
Sedimentibacter	0	2	2	0	1	0	0	0	**5**
Selenomonas	1	24	3	0	0	254	229	218	**729**
Skermanella	0	1	1	0	0	0	0	0	**2**
Smithella	1	3	7	0	0	0	0	0	**11**
*Sphingobacteriaceae*_unclassified	2	3	1	0	0	0	0	0	**6**
Sphingobacterium	0	1	1	0	0	0	0	0	**2**
Sphingomonas	6	7	10	0	0	0	0	0	**23**
Spiroplasma	1	0	0	0	0	0	0	0	**1**
Sporosarcina	0	1	0	0	0	0	0	0	**1**
Staphylococcus	13	15	5	0	0	0	0	0	**33**
Stenotrophomonas	7	15	11	0	1	0	0	0	**34**
Streptococcus	0	0	0	47	30	3	0	0	**80**
Sulfurospirillum	0	0	0	0	0	0	0	1	**1**
Symbiobacterium	1	0	0	0	0	0	0	0	**1**
*Synergistaceae*_unclassified	2	4	4	0	0	0	1	0	**11**
Syntrophomonas	0	2	2	0	0	0	0	0	**4**
*Syntrophorhabdus*_unclassified	0	0	1	0	0	0	0	0	**1**
Tepidiphilus	0	1	1	0	0	0	0	0	**2**
Terrisporobacter	1	1	0	0	0	0	0	0	**2**
Tetragenococcus	0	1	0	0	0	0	0	0	**1**
Thermoactinomyces	1	0	1	0	0	0	0	0	**2**
Thermus	1	2	2	0	0	0	0	0	**5**
Thiomicrospira	0	1	0	0	0	0	0	0	**1**
Tolumonas	0	0	0	0	0	2	3	5	**10**
Turicibacter	1	1	2	0	0	0	0	0	**4**
Ureibacillus	15	1	1	0	0	0	0	0	**17**
Uruburuella	0	0	0	1	0	0	0	0	**1**
Vagococcus	1	0	0	0	0	0	0	0	**1**

continued

continuation of Annex I

	C1	C2	C3	F1	F2	O1	O2	O3	TOTAL
Veillonella	0	1	0	0	0	13	5	14	**33**
*Veillonellaceae*_unclassified	1	12	0	0	0	128	126	115	**382**
*Verrucomicrobia*_unclassified	2	3	5	0	0	0	0	0	**10**
Vibrionimonas	0	0	1	0	0	0	0	0	**1**
Weissella	24	10	12	12	2	0	0	1	**61**
Wohlfahrtiimonas	1	1	1	0	0	0	0	0	**3**
*Xanthomonadaceae*_unclassified	0	0	0	0	0	0	1	1	**2**
Xanthomonas	2	0	2	0	0	0	0	0	**4**